Damaged Building Object Detection From Bitemporal Remote Sensing Imagery: A Cross-Task Integration Network and Five Datasets

Haiming Zhang, Yongxian Zhang, Di Wang, and Guorui Ma

***Abstract*— Existing change detection (CD) research mostly focuses on pixel-level dense prediction, while object detection (OD) for entire damaged/changed buildings is rare. Object-level detection datasets for damaged/changed buildings are also lacking. This article proposes an object-oriented damaged/changed building CD model, OoCDNet, along with five global-scale OD datasets for damaged/changed buildings. OoCDNet bridges and integrates the dual tasks of CD and OD, driven by the task of locating damaged/changed buildings. By modeling the changes in buildings at the object level between bitemporal images, it achieves rapid identification of target buildings. OoCDNet consists of four parts, the two paths of the dual-path feature extraction module (DouBackbone) are responsible for extracting the base features of the image, the bidirectional pyramid feature aggregation module (AggNeck) models the change information of the features aggregated at the end, the cross-informative self-attentive short-circuit enhancement module enhances the features from the DouBackbone with highly efficient self-attentive information and supplements the enhancement information to the AggNeck, and the enhanced features with semantic and localization information are fed into the detection module for OD. The proposed OoEWEBD is a global-scale OD dataset for damaged buildings, containing 10 377 image pairs, each sized 256 × 256 pixels. The remaining four datasets are created based on the WHU_CD, LEVIR-CD+, S2Looking, and xBD datasets, targeting buildings with general changes or those affected by disasters. Compared with state-of-the-art (SOTA) OD and CD methods, OoCDNet can quickly and effectively detect target buildings, achieving the highest accuracy and having a high application value. The code and datasets will be available at https://github.com/Haiming-Z/OD-based-Change-Detection-mode-OoCDNet.**

***Index Terms*— Building, change detection (CD), damage, object detection (OD), remote sensing.**

I. Introduction

CHANGE detection (CD) is based on bitemporal or multitemporal remote sensing images and analyzed by relevant algorithms and models to derive the change information of the region of interest or object of interest [1], [2]. Due to the practicality and considerable industry value of CD technology, CD has been applied and researched as a major scientific tool in many fields, such as land use assessment and analysis, urban planning and design, forest vegetation change monitoring, disaster damage assessment, global ecosystem monitoring, and even human social development analysis. Building CD (BCD) is a CD that takes buildings as the object of study, aiming to identify the location of buildings that have undergone ordinary changes or damage from remote sensing images [3]. With the successive launch of high-resolution (HR) optical remote sensing satellites around the world, the acquisition and use of HR remote sensing images have been gradually popularized [3], [4], and HR remote sensing images have become the main data source for BCD because they can provide richer texture details and contour information of buildings.

Almost all current BCD studies are based on dense prediction (DP-based), and few are based on object detection (OD-based). DP-based BCD uses bitemporal HR remote sensing imagery as input for the purpose of generating pixel-level building change masks. This mode of operation is well suited for the fine-grained assessment of building changes, such as monitoring the construction and demolition of buildings, analyzing the overall development of cities as well as villages, and assessing the damage of disaster-stricken buildings [5], [6]. However, this working mode is not friendly to some rapid census-type tasks and emergency-type tasks. For example, fast counting of damaged buildings in disaster-stricken areas, dynamic monitoring of unauthorized building construction in cities, and fast identification of house collapses in the context of war. This is because these tasks only focus on whether the building object as a whole has been damaged or changed and do not focus on the pixel-level mask of the change/damage information. It is difficult to determine the location of changed or damaged buildings because of the change mask with unclear or even incomplete boundaries. In addition, the time cost of dense prediction at the pixel level is often high. As shown in Fig. 1, both common algorithms and deep-learning-based DP-based BCDs need to go through the prediction pixel by pixel and obtain the binary mask map by determining whether each pixel belongs to a changing class or not. Furthermore, the commonly used end-to-end deep learning network architecture, on the other hand, needs to accomplish both feature

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downsampling and image recovery upsampling [7]. Moreover, deep learning methods rely heavily on high-quality pixel-level labeling labels. The process of pixel-level labeling is labor-intensive and much more complex compared to box-selected labeling for object-level labeling. Currently, there are a large number of public datasets available for research on DP-based BCD, whereas there is a lack of data benchmarks for research on OD-based BCD.

OD is also an important research direction in the field of remote sensing, which can be based on remote sensing images to calibrate the class of the object of interest and its rectangular boundaries [8], [9]. OD plays an important role in fields such as precision agriculture, disaster emergency response, urban planning, and military [9], [10]. Since HR remote sensing images contain a large amount of high-value ground observation information [8], efficient and effective utilization of such information is a requirement in the context of the massive growth of remote sensing image data. The current mainstream "one-step" OD method is capable of intensively predicting deep image features to achieve fast object discovery [11], which is very suitable for targeted research on objects of interest. As far as BCD is concerned, being able to quickly calibrate the rectangular area of changing buildings based on bitemporal HR remote sensing images has been able to meet the task requirements in many fields. According to our research, OD-based BCD is more advantageous than the DP-based BCD method in terms of work complexity and model efficiency. As shown in Fig. 1, OD-based BCD only needs to accomplish the determination of the object's rectangular coordinate positions without the complex pixel-by-pixel prediction process. Especially for deep learning methods, the OD-based BCD image modeling and the result generation process are more efficient than DP-based BCD. In addition, common remote sensing ODs are almost always single-image-based studies using publicly available datasets, such as DOTA [12], DIOR [13], and NWPU VHR [14], and there are almost no OD-based BCD studies based on two images. Although there has been extensive work on extracting buildings from single images [15], [16], these studies are still DP-based and are not specific to changed or damaged buildings. Therefore, we tried to develop a new BCD model, i.e., to understand changes from the OD perspective and perform OD-based BCD work.

Deep learning techniques have now become mainstream methods in remote sensing image processing [3], such as scene classification [17], image segmentation [18], CD, OD, and object extraction. Deep learning network models can extract complex, abstract, and nonlinear semantic features layer by layer from HR remote sensing images [19] and then apply these features to the downstream task side. As shown in Fig. 1, the general deep learning DP-based BCD method takes the bitemporal remote sensing image as input and realizes the feature extraction of the image through the encoding path and the decoding path restores the image size and completes the pixel-by-pixel prediction [20]. The time-consuming process of recovering the image size by layer-by-layer upsampling contributes to the inefficiency of such methods. In addition, skip connections, channel or spatial attention mechanisms [21], self-attention mechanisms [21], or other more complex network structures further increase the computational cost of the network. Worse still, pixel-level predictions tend to produce the inevitable pepper-and-salt phenomenon. In contrast, our proposed OD-based BCD model understands changes from the object's point of view and realizes CD by detecting the position of the rectangular box of the changing building, which, in turn, can avoid the effect of the pretzel noise completely. Furthermore, this model belongs to the mathematical regression problem, which is mainly based on highly abstract features for bounding box separation regression and category probability calculation. Moreover, the "one-step" OD method represented by YOLO [22], [23], [24], [25], [26] can quickly predict the object bounding box and category probabilities from the complete image, which is almost several times more efficient than the DP-based BCD method. Therefore, in order to realize a more effective and efficient BCD, we developed an object-oriented changed building detection model, OoCDNet. This is a new BCD model that enables the determination of change information at the object level of a building and accomplishes the detection of changed or damaged buildings. The specific innovations of this article are given as follows.

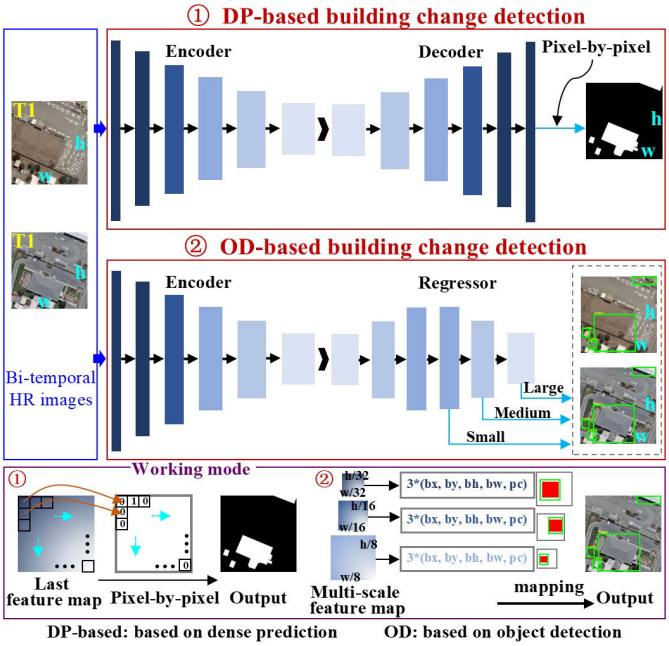


Fig. 1. Comparison of OD-based BCD and DP-based BCD working modes.

1) We propose a new BCD work model, OD-based BCD, which integrates the ideas and workflows of CD and OD and can improve the efficiency of BCD and serve applications in related fields.
2) We design the scene-level BCD network architecture, OoCDNet. Its four constituent modules are a complete whole for feature extraction, feature aggregation, feature enhancement, and object discovery. Cross-informative self-attentive short-circuit enhancement module CSCE-Block is a new lightweight self-attentive information enhancement module for spatial and semantic information enhancement of dual-temporal image features. Based on the deep semantic information self-supervision approach, it can complement low-dimensional features

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with stronger change information orientation at the network task execution end.

3) To fill the gap of damaged/changed building detection datasets and to provide benchmarks for OD-based BCD studies, we produce five new datasets: OoEWBD, OoWHU_CD, OoLEVIR-CD+, OoS2Looking, and OoxBD through extensive data collection and manual annotation. They can be used as OD-based BCD research benchmark datasets, providing data support for the practical application of BCD tasks in urban development assessment and emergency response.

II. Related Works

A. Change Detection

The development of CD methods has roughly gone through three stages [27], [28], [29]: the stage of traditional methods marked by algebraic computation and spatial transformation, the stage of machine learning methods marked by primary feature learning and clustering, and the stage of deep learning methods marked by deep feature extraction and modeling. Currently, deep learning-based AI methods dominate the field of CD research. From convolutional neural network (CNN)-based methods [30] to transformer-based methods [31], [32], Mamba-based methods [33], diffusion model-based methods [34], and foundational model-based methods, such as SAM [35] and SpectralGPT [36], various advanced CD approaches continue to emerge. Among them, CNNs are effective at capturing local features, with methods represented by fully convolutional networks [37] achieving significant results. However, due to the inherent limitations of CNNs' receptive field, transformer-based methods, which can establish long-range dependencies, have gradually taken the lead. In recent years, state-space models [38] have been applied as a viable alternative to transformers, particularly the more efficient Mamba architecture, which has shown strong competitiveness in many vision tasks, including CD. In addition, diffusion model-based methods, known for their exceptional ability to extract robust features and model complex training distributions [34], [39], have also been applied to CD tasks. Furthermore, using the strong generalization capabilities of large foundational models, such as SAM and SpectralGPT for CD, has demonstrated excellent performance. However, most general CD methods are interested in all elements within remote sensing imagery, detecting changes in various features and generating reports [40].

BCD is a type of CD that focuses on only one object, buildings. With the advancement of urbanization worldwide and the important role of buildings in the development of human society, the BCD method has gained more and more attention and applications. The development of BCD methods is roughly in the same vein as that of CD methods, and the first two phases of BCD methods can be divided into two categories: pixel-based methods and object-based methods [41], and most of the methods in the deep learning methods phase are pixel-based [42], and a small number of them are object-based [42]. Pixel-based methods mainly analyze the spectral difference information between pixels and use the threshold to obtain the change map, including algebraic operator method [43], change vector analysis [44], principal component analysis [45], support vector machine method [46], and random forest method [47], among others. Object-based methods can overcome the defects of pixel space discontinuity and the inability to take contextual information into account and obtain difference maps by segmenting pixels into independent objects and comparing their differences, including the object–object comparison method [40], the object comparison method after categorization [48], the simultaneous segmentation and categorization method [49], and so on. In addition, some strategies based on building index [50], image segmentation [51], mathematical morphology [51], and shadow detection [52] have been applied to BCD due to the special properties of buildings. Uncertainties associated with complex spectral and textural features in HR remote sensing images greatly limit the application of the first two phases of the method.

The current BCD is mainly based on DP-based deep learning methods. The general method works in an end-to-end mode, i.e., inputting a bitemporal remote sensing image and generating the same size binary mask. These methods can be categorized as pixel-based deep learning methods. There are also OD-based deep learning methods, such as ESCNet [53], ChangeOS [54], and S3Net [55]. Although these methods improve detection accuracy through deep learning techniques, their basic research units are still superpixels, rather than true OD-based research. The process of generating superpixels is often inefficient and limited in precision. End-to-end working models [56], more optimized network structures [27], effective attention mechanisms [21], [31], targeted loss functions [27], [57], and preprocessing and postprocessing strategies [58], [59] are different focuses of the iterative development of DP-based BCD methods. Regardless of the focus, the working model of these DP-based BCD methods remains unchanged, and the inherent drawbacks such as inefficiency, noise impact, and high training costs are difficult to overcome. However, the demand for change information among humans is becoming increasingly urgent. For example, some newly launched intelligent remote sensing satellites aim to provide real-time, accurate, and personalized remote sensing information services [60].

B. Object Detection

The development of OD methods can also be broadly categorized into three stages: the traditional method stage, the two-step detection stage, and the one-step detection stage. Traditional methods mostly use artificial features to detect specific objects, such as the mixed model method [61], the oriented gradient histogram method [62], and the Hough transform method [63]. These methods have low generalization performance because they only involve the low-level features of the object and are difficult to effectively describe the diversity and complexity of the object. It is even more difficult to apply them to HR remote sensing images. With the application of convolutional neural networks in the field of OD, two-step detection methods have been devised. Such methods can

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utilize deep neural networks to extract deep abstract features of the image, which can effectively obtain a high-level semantic feature representation of the target object.

R-CNN [64] is the earliest algorithm to apply deep learning to OD, which is based on CNN for feature extraction and utilizes support vector machines and regression algorithms for object class determination and bounding box inference. Furthermore, Fast R-CNN [65], Faster R-CNN [66], and Mask R-CNN [67] have been proposed successively. They have achieved further improvements in detection accuracy, efficiency, and multitask collaboration. However, the training and inference processes in the two-step detection method seriously affect the efficiency of this type of method due to the fact that RPN is unavoidable. The one-step detection method does not use RPNs for initial detection but directly regresses the distribution probability and positional coordinates of the object and, thus, has been widely concerned and continuously researched. Such methods are represented by the YOLO family of networks, including YOLOv1 [22], YOLOv2 [68], YOLOv3 [69], YOLOv4 [23], YOLOv5 [70], YOLOv6 [25], YOLOv7 [26], YOLOv8, YOLOv9 [71], and YOLOV10 [72]. There are also some task-specific variants, such as SCRDet [73], R3Det [74], and FANet [9], among others. After continuous and iterative development, the one-step detection method has achieved satisfactory accuracy and efficiency and has become the dominant method in the field of OD at present. However, there are hardly any attempts to conduct CD studies using OD methods. All of the above methods input one image and generate detection results. When the input is two images and there are variations between the images, it poses a challenge to the OD method. It may be a challenge to understand the change from an OD perspective for studying special objects such as buildings.

C. Problem Statement and Motivation

To support emergency response and urban development assessment, it is necessary to investigate damaged or changed buildings. Emergency response primarily involves buildings damaged due to war, conflict, explosions, and other similar contexts, while urban development assessment focuses mainly on newly constructed or demolished buildings within cities. Currently, different countries or regions adopt various measures for building investigations, which largely depend on their policies, technological capabilities, and census objectives. For example, the United Nations Office for the Coordination of Humanitarian Affairs (OCHA) focuses on on-site search, rescue, and coordination after disasters, collecting and analyzing the impact following a disaster or conflict (such as location information of damaged buildings), which is essential for the organization to formulate plans and strategies [75]. Meanwhile, China's Urban Growth Boundary Policy [76] primarily focuses on changes in buildings near development boundaries to limit unnecessary urban sprawl and inefficient expansion. Although the development of both CD and OD methods has made great progress, they remain independent. Some CD or OD methods are not part of the OD-based BCD research even though they enable the detection of damaged/changed buildings in certain areas. Furthermore, with the demand for mass and real-time remote sensing image processing, especially with the current development of intelligent remote sensing satellites [77], the acquisition, processing, and application modes of remote sensing data have changed dramatically. The traditional satellite remote sensing system needs to go through a complex and lengthy process of four mutually independent links, namely, operation and control, reception, processing, and application, with a slow response time that cannot meet emergency needs and users' high time-sensitive needs. As shown in Fig. 2, intelligent remote sensing satellites can support on-orbit processing and real-time service of data, and end-to-end real-time transmission can be realized from satellites to users. This, in turn, serves many issues of international concern, such as urban development assessment, emergency response, and humanitarian assistance. However, to guarantee the mass service of remote sensing images and to realize higher emergency response efficiency, it is necessary to realize the cooperative work of the tasks of location sensing, object sensing, and change sensing at the algorithmic level. A CD method that takes into account object location, object type, and object change is necessary to realize satellite-based processing of remote sensing tasks. The high time-consuming DP-based CD method requires strictly aligned bitemporal images, and the up-front alignment process will undoubtedly increase the burden of on-planet processing. The pure OD method, which is not adapted to change perception, cannot perform the CD task. In order to adapt to the demand for fast, accurate, and flexible production of remote sensing information products and to meet the multifaceted needs for urban development assessment, rapid investigation of global natural disasters, international humanitarian assistance, emergency response to emergencies, and so on, the study of OD-based fast CD methods is extremely valuable.

Buildings are almost the most important man-made features and typical targets on the ground, and the OD method is the optimal means to perceive the targets. By integrating the design concept of CD, the natural advantages of the OD method can be effectively utilized to realize the "one-step" fast query of damaged/changed buildings. In contrast, even the lightweight CD method requires a large number of pixel-level predictions, and the postprocessing process can limit its application. Therefore, we understand the change from the perspective of the object and achieve the effective perception of building objectives through the OD-based model (network model with datasets) with fast, efficient, and precise task orientation.

III. Object-Oriented Changed Building Detection Datasets

A. Global-Scale Damaged Building Detection Dataset

To provide a data baseline for OD-based BCD studies of damaged buildings and to take into account the causes of building damage (natural or man-made), we collected bitemporal images of damaged buildings in the context of earthquakes and wars. Table I details the location of the images, the time they were taken, and other information. The distribution of the data can be seen in Fig. 3. The OD-based labeling of

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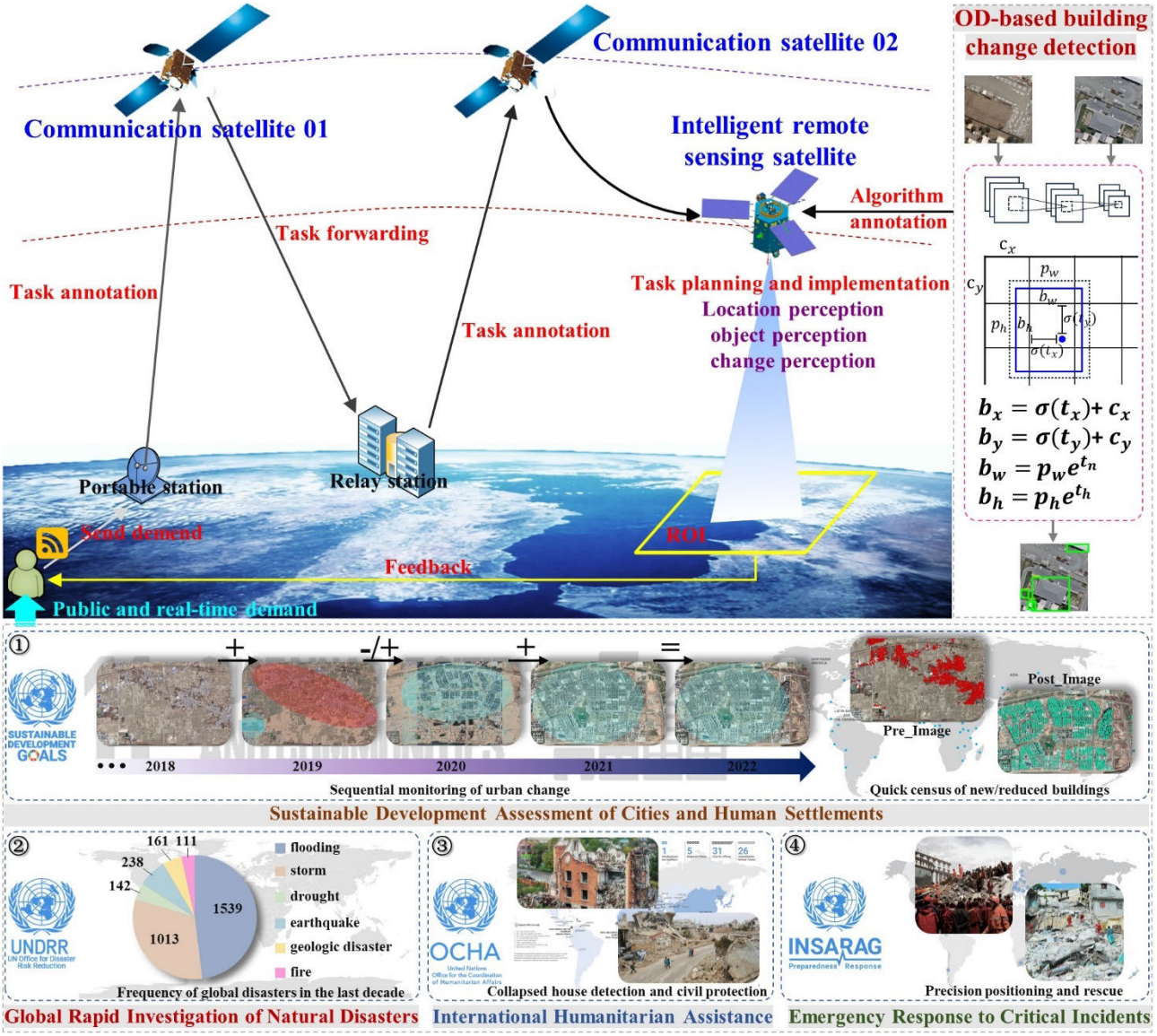


Fig. 2. Design concept and application future of OD-based BCD.

these images was completed through visual interpretation by professionals. A total of 10 377 samples of 256 × 256 pixel size were obtained, containing 52 661 instances of damaged buildings.

B. Object-Oriented Change Detection Datasets

As shown in Fig. 4, we obtained OD-based labels based on the DP-based labeling in the WHU_CD [78], LEVIR-CD+ [79], S2Looking [80], and xBD [81] datasets. Each patch sample in OoWHU_CD, OoLEVIR-CD+, OoS2Looking, and OoxBD contains prechange and postchange images and OD-based labels. The size of the image is 256 × 256 pixels. The four datasets, respectively, contain 7680, 15 760, 80 000, and 22 068 samples, with 3281, 31 333, 65 920, and 850 736 instances of changed or disaster-affected buildings.

The WHU_CD and LEVIR-CD+ datasets contain imagery that covers mainly urban areas, while the S2Looking imagery covers multiple rural areas around the globe. These three datasets are mainly concerned with the growth (new construction of buildings) and decline (demolition of buildings) of buildings. The xBD dataset consists of images from 19 different disaster events worldwide, primarily involving buildings affected by earthquakes, floods, wildfires, etc.

The images in the WHU_CD and LEVIR-CD+ datasets have undergone strict geometric correction, while the images in the S2Looking dataset were captured using satellite rolling imaging. The images in the xBD dataset have higher scene complexity. Therefore, these datasets provide diverse benchmarks for algorithm testing. A reduced version of our dataset has been published: https://github.com/Haiming-Z/OD-based-BCD-datasets.

IV. Methodology

A. Overview of OoCDNet

The overall structure of OoCDNet is shown in Fig. 5. The inputs to the network are independent bitemporal HR remote sensing images, and the two paths of the double-path feature extraction module (DouBackbone) receive the T1

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TABLE I
OVERVIEW OF THE OOEWBD DATASET

Sites	Times	Descriptions	Images	Sources	Samples	Instances
Haiti	2010.1.12	The capital, Port-au-Prince, was severely affected by the 7.3 magnitude earthquake. Urban areas.	83 pairs, 0.3m resolution	Google Earth	407	797
Lebanon	2020.8.4	A huge explosion in Beirut harbor damaged several houses. Urban areas.	1 pair, 0.3m resolution	Maxar	76	132
Turkey	2023.2.6	The 7.8 magnitude earthquake flattened many residential buildings in the neighborhood. Urban and rural areas.	13 pairs, 0.3m resolution	Google Earth	1386	2749
Ukraine	2022.2.25	A large number of buildings in the Mariupol area were hit by shells and suffered various degrees of damage. Urban areas.	25 pairs, 0.3m resolution	Google Earth	3842	6771
Syria	2016.10.21	The Omar oil field was significantly damaged in the air strikes. Urban areas.	4 pairs, 0.3m resolution	Google Earth	38	67
Nepal	2015.4.25	The 8.1 magnitude earthquake destroyed hundreds of thousands of buildings. Rural areas.	121 pairs, 0.3m resolution	Google Earth	719	3843
China	2010.4.14	The 7.1 magnitude earthquake collapsed tens of thousands of homes in Yushu. Urban and rural areas.	39 pairs, 0.5m resolution	Google Earth	412	5864
China	2015.8.12	A huge explosion in Tianjin's Binhai New Area damaged more than 300 houses. Urban areas.	1 pair, 0.3m resolution	Google Earth	109	240
Indonesia	2018.9.28	Thousands of houses were damaged by the 7.4 magnitude earthquake. Urban and rural areas.	48 pairs, 0.3m resolution	Google Earth	784	5353
Japan	2011.3.11	The 9.0 magnitude earthquake, which triggered a tsunami, caused devastation in Iwate Prefecture and other areas. Urban and rural areas.	138 pairs, 0.5m resolution	Google Earth	2604	26845

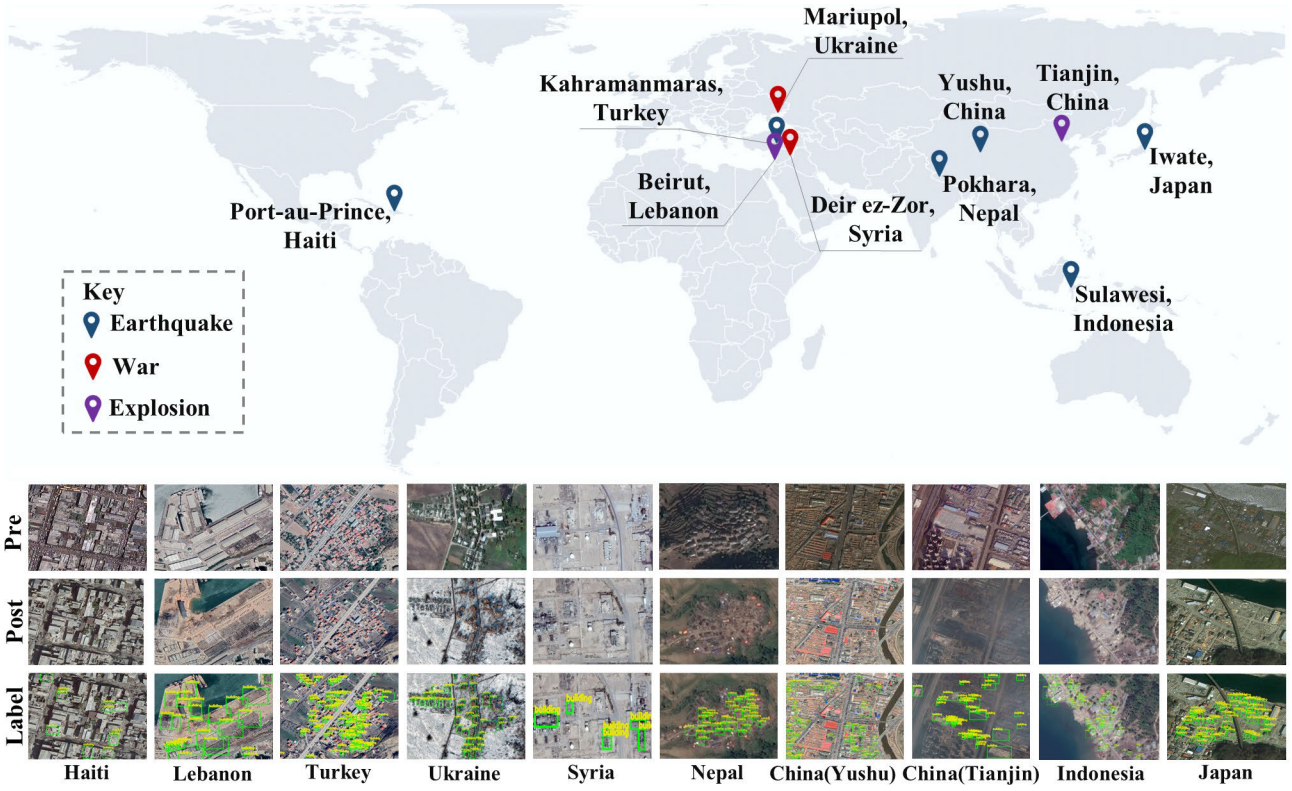


Fig. 3. Example of the OoEWBD dataset and distribution.

and T2 images, respectively, and undergo ten layers of feature encoding to fully extract the underlying features in the images. During this period, there are five downsampling operations to reduce the spatial resolution of the feature map to half of the original one, respectively. The dual-path structure uses a weight-sharing model to ensure the stability of the network encoding and to enable the network to model the change information consistently. The feature map is forced to be aggregated at the end of DouBackbone after a fast spatial pyramid pooling (SPPF) operation as a high-dimensional change feature set for the downstream task pipeline. The bidirectional pyramid feature aggregation module (AggNeck) processes the feature groups in two steps, upsampling and downsampling, to aggregate semantic and localization features to obtain task-oriented hierarchical features.

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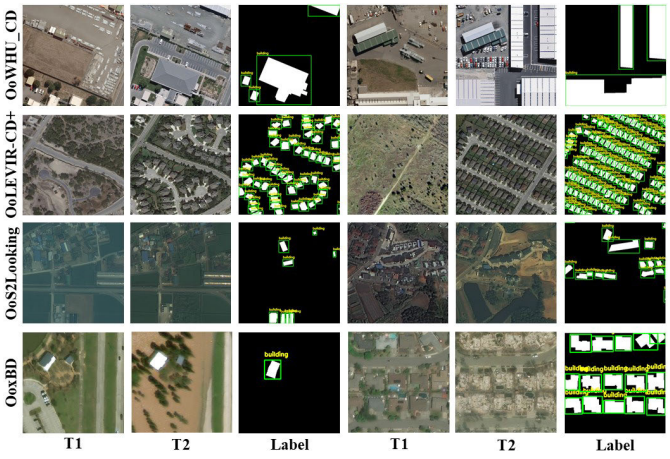


Fig. 4. Examples of an object-oriented change detection dataset.

To emphasize the differences between features, capture more representative change features, and ensure the efficiency of the model, we designed a cross-informative self-attentive short-circuit enhancement module, CSCEBlock. The features of the corresponding layers in the two paths of DouBackbone are processed for differencing and concatenation and are subsequently cross-fed into the CSCEBlock to receive supervision of higher level semantic feature information and global dependency enhancement. The augmented features have stronger localization and change semantic information and are supplemented into higher dimensional feature layers. Finally, the outputs of the three AggNeck feature layers are fed into the change building detection module (Detect) for the detection of different-sized objects.

B. Detail of OoCDNet Stages

1) Workflow Between Modules: Bitemporal HR remote sensing images are simultaneously subjected to Mosaic data enhancement operations to obtain training data with higher background complexity and more diverse images. The enhanced training data help improve the generalization ability of the model. After that, the image I_{T1} and the image I_{T2} of size, $H \times W$, are fed into the A and B paths of DouBackbone for feature extraction, respectively. In this process, the images are processed in six sets of modularity, and the convolution operation extracts the local feature representations layer by layer. Moreover, the semantic nature of the features describing the local changes will gradually increase as the number of layers deepens. Among them, Convblock is the standardized convolution module, which consists of a large kernel convolution operation unit, a batch normalization processing unit, and an SiLU activation function. The first group of processing modules contains only Convblock, which is responsible for feature space conversion and dimension enhancement operations on the image, which can convert I_{T1} and I_{T2} from the color space to the feature space to obtain the base feature output $f_{D_1}^{C} \in \mathbb{R}^{2^{-1}H \times 2^{-1}W}$. The calculation formula is given as follows:

$$f_{D_1}^{C} = \sigma(\mathcal{B}(\mathrm{Conv}(I_{T1/T2})) \tag{1}$$

where Conv is the convolution operation with a convolution kernel size of 7, B is the batch normalization operation, and σ is the SiLU activation function.

Groups 2–5 contain two processing modules, Convblock and ResConvblock, in the processing stage, which can extract feature representations $f_{D_l}(l \in [2, 3, 4, 5])$ with a higher degree of abstraction layer by layer. Each group of Convblock is responsible for scaling and dimensioning the feature map to get the output $f_{D_l}^{C_l} \in \mathbb{R}^{2^{-l}H \times 2^{-l}W}$. ResConvblock is the residual structure, which is responsible for the deep feature modeling, and uses a small convolution kernel to understand the interimage changes step-by-step to get the output $f_{D_l}^{R_l} \in \mathbb{R}^{2^{-l}H \times 2^{-l}W}$. The combination of convolution operation and residual operation enables the full extraction of features. The calculation formula is given as follows:

$$f_{D_l}^{R_l} = \mathrm{Conv}([\mathrm{Conv}(f); \Re(\mathrm{Conv}(f))]) \tag{2}$$

where f is f_{D_l} for each layer, Conv is the mapping operation with convolution kernel size 1, and R is the residual operation.

DouBackbone ends with SPPF, which is responsible for the pyramid pooling of high-level abstract features that have been deeply compressed and merging feature representations at different scales of the same feature map to obtain the multiscale fusion feature $f_{D_6}^{S}$. The fusion feature has stronger stability and is a reliable feature to be used as a detection task. The calculation formula is given as follows:

$$f_{D_6}^{S} = \mathrm{Conv}([\mathrm{Conv}(f); \mathcal{M}(f'); \mathcal{M}(f''); \mathcal{M}(f''')]) \tag{3}$$

where f is $f_{D_l}^{5}$, $\mathcal{M}$ is the maximum or average pooling operation, and $f' \sim f'''$ is the same-scale pooling feature map.

$f_{D_6}^{S}$'s of the two paths A and B are concatenated to obtain the base feature output $f_{D_6}^{S_c}$, which is used as input to AggNeck. In AggNeck, there are two paths, C and D, for feature tasking mapping. The basic constituent modules of each path are also Convblock and ResConvblock. The converged features in path C are first dimensionally compressed by the Convblock module with convolutional kernel size 1 to get the feature output $f_{A_1}^{C}$ with dimension 256. Afterward, the size of $f_{A_1}^{C}$ is scaled up twice using the UpSample operation, and feature reabstraction is performed to obtain $f_{A_2}^{C}$ within the ResConvblock and Convblock processing modules, in turn. Finally, the feature map is zoomed in twice after one more UpSample operation. The features ($f_{A_2}^{U}$ and $f_{A_3}^{U}$) obtained after each UpSample operation are first fed into the CSCEBlock for semantic self-attention enhancement of the shallow features and later merged with the enhanced features f_{C_l} from the CSCEBlock to complement the global dependency information. f_{C_1} from layer ① is merged with the f_{C_3} from layer ③ and added to the features after the second UpSample operation to obtain $f_{A_3}^{C_c}$, i.e., $f_{A_3}^{C}$. The ones from layers ② and ④ are then added to the features after the first UpSample operation to obtain $f_{A_2}^{C}$. In AggNeck's D-path, two more downsampling operations are performed based on $f_{A_3}^{C}$, and the feature maps after supplementing the strong localization information are secondarily abstracted to obtain the outputs of the hierarchical task features with strong semantic representations, $f_{A_1}^{D}$, $f_{A_2}^{D}$, and $f_{A_3}^{D}$. To supplement the strong localization information

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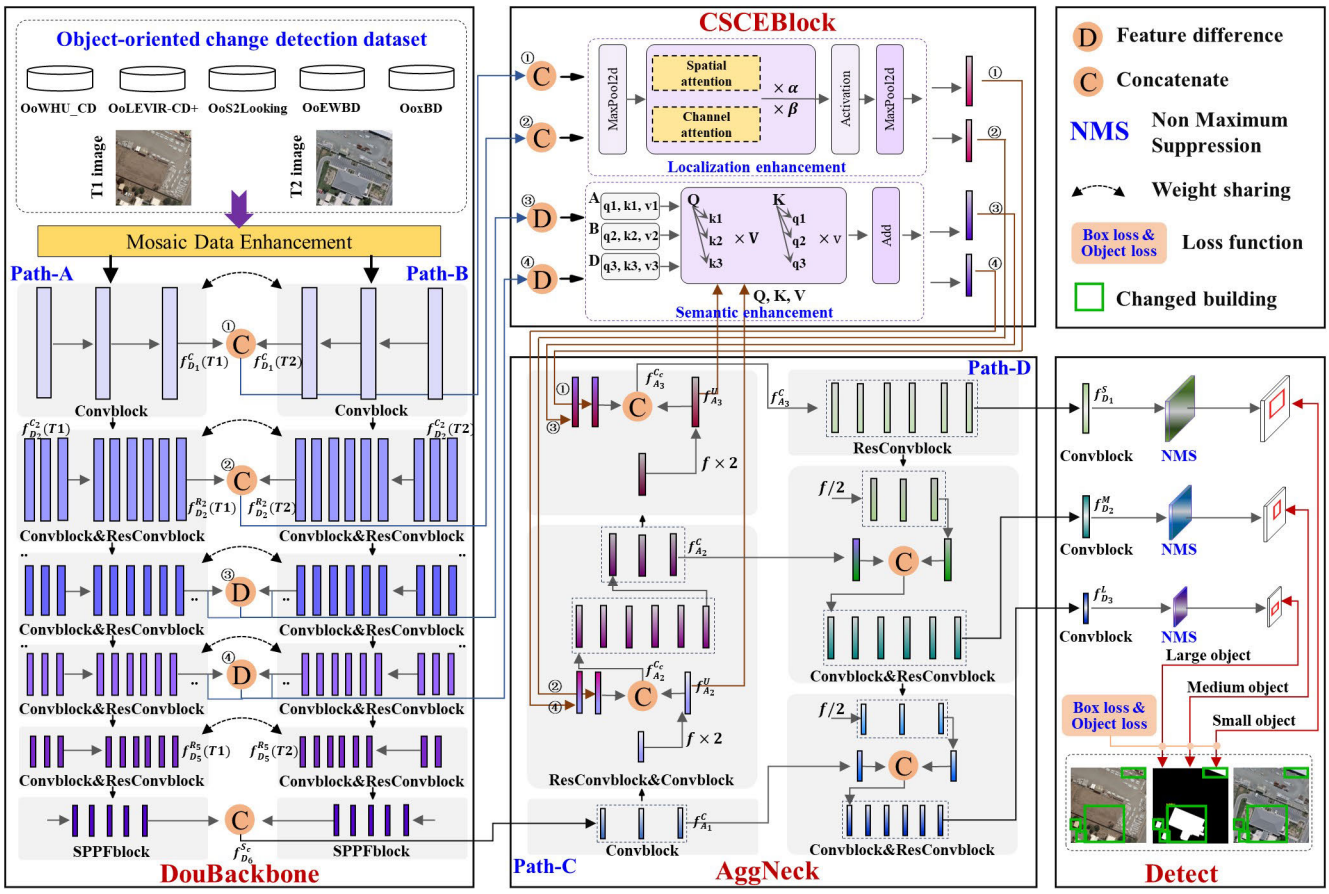


Fig. 5. Schematic of the OoCDNet network structure.

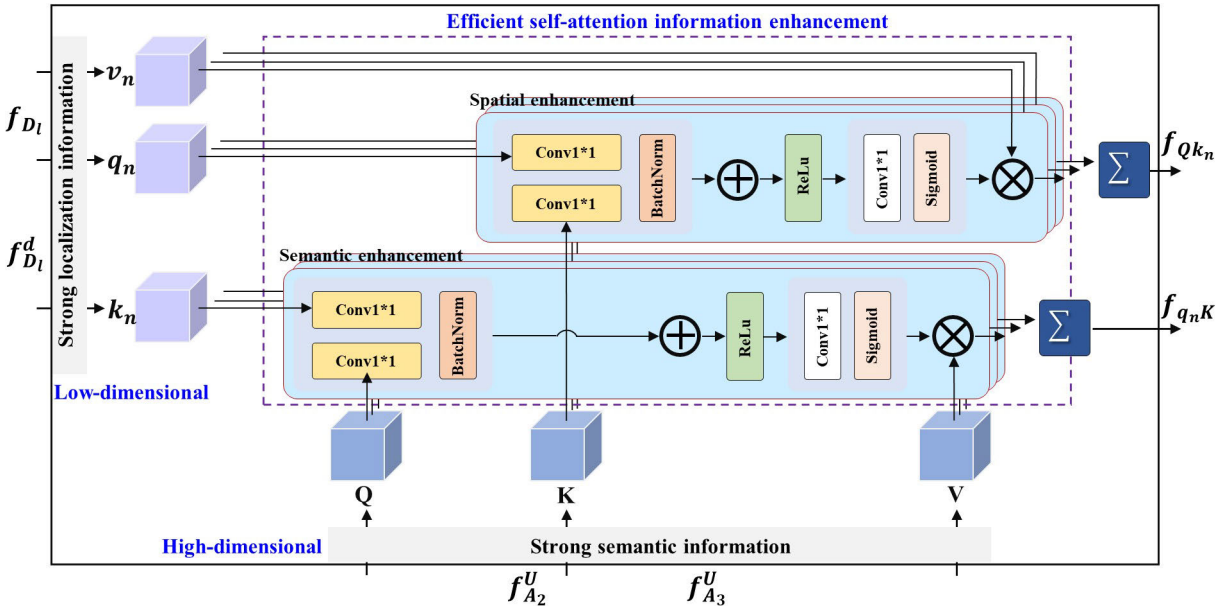


Fig. 6. CPA self-attention enhancement module.

again, $f_{A_1}^C$ and $f_{A_2}^C$ are merged with the features after the two downsampling operations, respectively, and are fed into the Convblock and ResConvblock modules for processing. Eventually, $f_{A_1}^D$, $f_{A_2}^D$, and $f_{A_3}^D$ will be used as feature inputs before performing the detection task.

In detect, $f_{A_1}^D$, $f_{A_2}^D$, and $f_{A_3}^D$ are subjected to a mapping operation with a convolutional kernel size of 1 followed by nonlinear activation of the Sigmoid function to obtain the outputs $f_{D_3}^L$, $f_{D_2}^M$, and $f_{D_1}^S$, which were used to predict the detection boxes and the confidence levels. $f_{D_3}^L$, $f_{D_2}^M$, and

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$f_{D_1}^S$ are used to predict large, medium, and small objects, respectively. Finally, the output is obtained after NMS filtering. The final output of the network, f_D, is calculated as follows:

$$f_D = \mathrm{NMS}\left(f_{D_3}^L\right) \cup \mathrm{NMS}\left(f_{D_2}^M\right) \cup \mathrm{NMS}\left(f_{D_1}^S\right) \tag{4}$$

where $\cup$ denotes the result merge operation and NMS denotes the nonmaximum suppression operation.

2) CSCEBlock-Based Self-Attentive Cross-Information Enhancement: The DP-based CD method adequately extracts the change features in the bitemporal image through the encoding and decoding structure and models the interimage change information, and the end-to-end segmentation results in a task output that allows the model to learn the pixel-level change mapping relationship. However, the OD-based CD method determines the location coordinates of the object employing regression, and highly abstract features are used as predictive values in the regression process. The changing relationship between bitemporal images can hardly be learned during the regression process. The extraction of primary change features and the reliability of feature information are especially important for OD-based CD methods. In addition, the problem of primary feature information loss also occurs during the layer-by-layer abstraction of features. The problem of feature misalignment between two images also becomes more abstract with the increasing number of convolutional layers, which makes it difficult to handle. Therefore, to ensure the network's suitability for the CD task, to improve the network's ability to capture and model the changing information, and to obtain the high-dimensional feature information that is more valuable for the OD-based CD task, we designed the CSCEBlock.

CSCEBlock is a lightweight deep semantic feature-supervised self-attention mechanism, which can receive low-dimensional features from the dual-path encoding layer and high-dimensional features from the deep abstraction layer, supplement the high- and low-dimensional features into the deep abstraction paths after strengthening the information on the spatial and semantic levels, provide high-value and high-reliable feature information for the OD-based CD task, and effectively solve the problem of deep feature misalignment. In addition, to mitigate the problem of primary feature loss after the image is highly abstracted, CSCEBlock can also supplement the primary features from the shallower layers within DouBackbone into AggNeck. The low-dimensional features $f_{D_l}(l \in [1,2])$ from layers ①–④ of the A and B paths and the high-dimensional features $f_{D_l}(l \in [3,4])$ from the two upsampling operations of the C path are fed into the CSCEBlock for feature enhancement, and the outputs f_{C_1}, f_{C_2}, f_{C_3}, and f_{C_4} are obtained. Features in layers ① and ② have larger scales and stronger localization information, and they execute channel and spatial attentional enhancement. The features in layers ③ and ④ have smaller scales and rich semantic information, and they execute deep semantic self-attention enhancement. The features of layers ① and ② are input into CSCEBlock after concatenation, respectively, and the feature map is first downsampled by a factor of 2, after which the channel and spatial attention mechanisms are used to enhance the feature map, respectively, and finally, the feature map is shrunk by a factor of 2 after one more downsampling to obtain the output f_{C_1} or f_{C_2}. The calculation formula is given as follows:

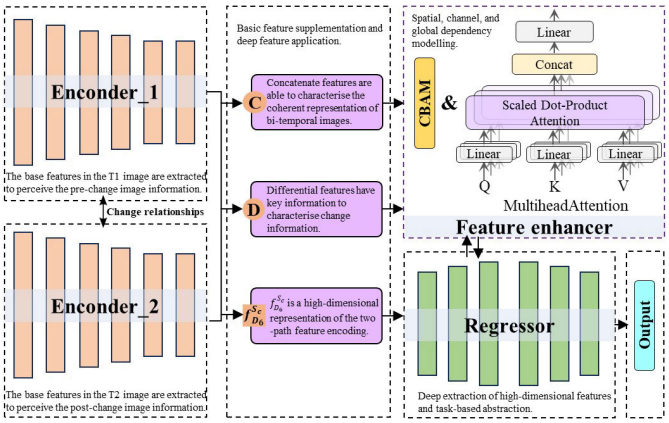


Fig. 7. Conceptual diagram of internal feature extraction and enhancement in OoCDNet.

$$f_{C_l} = \mathrm{Down}(\mathrm{Spa}(\mathrm{Cha}(\mathrm{Down}(f_{D_l})))), \quad (l = 1, 2) \tag{5}$$

where Down denotes the downsampling operation, Spa denotes the spatial attention mechanism, and Cha denotes the channel attention mechanism.

The low-dimensional features from layers ③ and ④ contain f_{D_l} of the A and B paths and their different features $f_{D_l}^d(l \in [3,4])$, as well as the high-dimensional features from AggNeck, $f_{A_2}^U$ and $f_{A_3}^U$. All the low-dimensional features are mapped into the three sets of intermediate feature representations of q, k, and v by using the 1×1 convolution operation, respectively, and the high-dimensional features are mapped into three sets of intermediate feature representations of Q, K, and V. After mapping, all the intermediate features have the same number of channels, and they broaden the representation of the original low- or high-dimensional features. After that, a self-attentive query from semantic to spatial information is performed using Q with stronger semantic information with all k with stronger spatial information, and the attention coefficient matrix W_{Qk} is computed. W_{Qk} is then multiplied with v to get the output $f_{Qk_n}, (n \in [1,2,3])$, which achieves spatial information enhancement for high-dimensional strong semantic features. Similarly, a self-attentive query from spatial to semantic using all q and K is computed to obtain the attention coefficient matrix W_{qK}. W_{qK} is multiplied with all v to obtain the output $f_{q_nK}, (n \in [1,2,3])$, which achieves semantic information enhancement for low-dimensional strong spatial features. All f_{Qk_n} and f_{q_nK} are, respectively, summed within groups to obtain aggregated features before merging by channel and modeling interchannel relationships using channel attention. Finally, the final output $f_{C_l}, (l \in [3,4])$ is obtained by normalization and nonlinear activation. The calculation formula is given as follows:

$$f_{Qk_n} = \mathrm{CPA}(Q, k, V), \quad (l = 1, 2, 3) \tag{6}$$

$$f_{q_nK} = \mathrm{CPA}(q, K, v), \quad (l = 1, 2, 3) \tag{7}$$

$$f'_{C_l} = \mathrm{Cha}\left(\mathrm{cat}\left(\sum f_{Qk_n} + \sum f_{q_nK}\right)\right) * \mathrm{cat}\left(\sum f_{Qk_n} + \sum f_{q_nK}\right) \tag{8}$$

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$$f_{C_l} = \sigma\left(\mathcal{B}\left(f'_{C_l}\right)\right) \tag{9}$$

where CPA denotes the self-attentive computational module, $Q, K, V, q, k, v \in \mathbb{R}^{h\times w}$, $h = 2^{-3/4}H$, $w = 2^{-3/4}W$, and $\sum(\cdot)$ denotes the feature map accumulation operation.

As shown in Fig. 6, within the CPA, the attention coefficient matrix is first computed via Q and k or q and K. The matrix is then mapped to the attention coefficients. The mapped intermediate features are different representations of spatial and semantic information, and by summing them and going through nonlinear activation, one can implicitly model spatial information based on semantic information or semantic information based on spatial information. After that, then the features can be aligned by mapping operation, and through the activation function, the final attention coefficient matrix is obtained. V or v is multiplied with the attention coefficient matrix to achieve the enhancement of the features. In addition, the outputs of layers ① and ③, ②, and ④ are cross-fed into AggNeck, which can be hardened to achieve further enhancement of spatial and semantic information, as well as the supplementation of low-dimensional feature information.

3) Loss Function Configuration: The OD task has three purposes: to detect the position of the object in the image, to detect the size of the object, and to classify the detected object. Since CD involves only change and unchanged, OD-based CD is a single classification task and does not involve the problem of classifying the object. The metrics for the first two purposes are the loss of confidence (Loss_{obj}) and the loss of the rectangular box (Loss_{box}). Among them, Loss_{obj} is responsible for calculating the confidence of the network, which measures the accuracy of the network's judgment of the object, and is calculated using BCE logits loss. Loss_{box} is responsible for calculating the error between the prediction box and the calibration box, which measures the size of the overlap between the prediction result and the real value, and is calculated using GIoU loss. The composition of the overall loss function contains only these two parts, defined as follows:

$$\text{Loss} = \lambda\text{Loss}_{\text{obj}} + \mu\text{Loss}_{\text{box}} \tag{10}$$

$$\text{Loss}_{\text{obj}} = \frac{1}{N}\sum_{i=1}^{N}\left(y_i \log\left(\sigma(p_i)\right) + (1-y_i)\log\left(1-\sigma(p_i)\right)\right) \tag{11}$$

$$\text{Loss}_{\text{box}} = 1 - \left(\frac{|A\cap B|}{|A\cup B|} - \frac{|C(A\cup B)|}{|C|}\right) \tag{12}$$

where λ and μ are loss term coefficients, N is the number of pixels, y_i is the label, p_i is the prediction probability, σ is the sigmoid function, A is the prediction box, B is the calibration box, and C is the small enclosing convex. To improve the percentage of Loss_{box}, the values of λ and μ are taken as 1.0 and 0.5, respectively.

Corresponding to Algorithm 1, Fig. 7 illustrates the process of internal feature extraction and enhancement in OoCDNet. Overall, OoCDNet consists of three parts: dual encoders, a feature enhancer, and a regressor. OoCDNet features a dual-path feature encoding structure, allowing it to extract fundamental features that characterize changes from T1 and T2 images layer by layer. The weight-sharing mechanism helps the model in modeling the changes between the bitemporal images. In contrast, ordinary OD models almost all have a single-path encoding structure, which is not suitable for CD tasks. This is because the single-path encoding approach makes it difficult to extract key features from the concatenated bitemporal images, and the six-channel images are hard to feed into the model. To provide sufficient and effective fundamental features for the regression stage, the differential features and concatenated features between the two encoders are provided to the regressor.

Algorithm 1 shows the OoCDNet-based object-level detection method for changed buildings.

Meanwhile, to avoid the influence of low-level semantic errors in the fundamental features, high-level features in the regressor are introduced into the feature enhancer to enhance the fundamental features. In the feature enhancer, higher resolution features are enhanced and suppressed in both spatial and channel dimensions through CBAM operations. Lower resolution features are self-attention enhanced with the high-level features from the regressor, capturing key features by modeling global dependencies. The features from the enhancer are supplemented into the regressor and further compressed to obtain three layers of task-oriented features.

C. Evaluation Metrics

To objectively evaluate the performance and capability of OoCDNet, comparative analyses were conducted from two perspectives, OD and CD, respectively. For OD, Faster RCNN [66], SSD [82], YOLOv3 [69], M2Det [83], EfficientDet [84], DETR [85], YOLOv5, YOLOX [86], YOLOv7 [26], YOLOv8, YOLOv9 [71], YOLOv10 [72], R3Det [74], S2ANet [87], ORCNN [88], SASM [89], and ORepPoints [90] were used as the comparison method. The R3Det, S2ANet, ORCNN, SASM, and ORepPoints are OD models designed for remote sensing imagery. They are considered advanced and classic methods and have now been integrated into the MMRotate framework, commonly used as benchmarks for OD approaches. The other models are general OD methods, serving as typical models in the field of OD and representing its development trajectory. In addition, nearly all newly proposed OD models are compared against several of these models. For CD, Cosegmentation [91], S3Net [55], UNet [20], FC_EF_Res [92], DeepLabv3+ [93], and BIT [32] were used as comparison methods. It is worth mentioning that the comparison with the OD method is to demonstrate the effectiveness and modeling accuracy of OoCDNet for the discovery of changing buildings, and the comparison with the CD method is to demonstrate the feasibility of OD-based CD. The quantitative evaluation metrics used for the comparison with the OD method are precision (P), recall (R), and mean average precision (mAP_{50}, $\text{mAP}_{50\text{–}95}$). The quantitative evaluation metrics used to compare with the CD method are P, R, $F1$ value, and intersection over union (IoU), with the following equations:

$$P = \text{TP}/(\text{TP}+\text{FP}) \tag{13}$$

$$R = \text{TP}/(\text{TP}+\text{FN}) \tag{14}$$

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Algorithm 1 Changed Building OD With OoCDNet

Input: The training dataset D_M containing image pairs (I_{T1}, I_{T2}) and labels La, M (the number of datasets)
Output: OoCDNet models
Procedure Training

$D_M^E \leftarrow$Data enhancement of (D_M, La) using mosaic strategy
$OoCDNet^0 \leftarrow Warmup(I_{T1}, I_{T2})$
while epoch $\leq$ max_epoch **do**
  for B $(I_{T1}, I_{T2}, la) \in D_M^E$ **do**
    DouBackbone stage:
      $f_{D_1}^C(T1/T2) \leftarrow \sigma(\mathcal{B}(Conv(I_{T1/T2})))$
      $f_{D_l}^{C_l}(T1/T2) \leftarrow Convblock(f_{D_1}^C)$
      $f_{D_l}^{R_l}(T1/T2) \leftarrow Conv([Conv(f_{D_l}^{C_l}); \Re(Conv(f_{D_l}^{C_l}))])$
      $f_{D_6}^S(T1/T2) \leftarrow SPPF(f_{D_l}^{R_l})$
      $f_{D_6}^S \leftarrow Cat(f_{D_6}^S(T1), f_{D_6}^S(T2))$
    AggNeck stage:
      $f_{A_1}^C \leftarrow Convblock(f_{D_6}^S)$
      $f_{A_2}^U \leftarrow UpSample(f_{A_1}^C)$
      $f_{C_l} \leftarrow CSCEBlock(f_{A_2}^U, f_{D_l}^d)$
    if l = 1 **then**
      $f_{A_3}^C \leftarrow cat(f_{C_1}, f_{C_3})$
    end if
    if $l = 2$ **then**
      $f_{A_2}^C \leftarrow cat(f_{C_2}, f_{C_4})$
    end if
    $f_{A_1}^D, f_{A_2}^D, f_{A_3}^D \leftarrow Path_D(f_{A_3}^C, f_{A_1}^C, f_{A_2}^C)$
    CSCEBlock stage: (Simultaneous with AggNeck)
    if l = 1 or 2 **then**
      $f_{C_l} \leftarrow$Spatial and channel attention enhancement (f_{D_l})
    end if
    if l = 3 or 4 **then**
      $f_{C_l} \leftarrow$Self attention enhancement $(f_{D_l}, f_{D_l}^d, f_{A_2}^U, f_{A_3}^U)$
    end if
    Detect stage:
      $f_D \leftarrow NMS(f_{D_3}^L, f_{D_2}^M, f_{D_1}^S)$
    Supervision: (The coefficients λ and μ)
      $Loss_{\text{obj}} \leftarrow BCE(y_obj, l)$
      $Loss_{\text{box}} \leftarrow GIoU(y_box, l)$
      $L \leftarrow \lambda Loss_{\text{obj}} + \mu Loss_{\text{box}}$
  end for
end while
end procedure:

$$\text{mAP} = \frac{1}{c}\sum_{i=1}^{c}\left(\int_0^1 P_i(R)dR\right) \tag{15}$$

$$F1 = 2\text{TP}/(2\text{TP} + \text{FP} + \text{FN}) \tag{16}$$

$$\text{IOU} = \text{TP}/(\text{TP} + \text{FP} + \text{FN}) \tag{17}$$

where TP is the number of correctly categorized positive samples, FP is the number of incorrectly categorized negative samples, FN is the number of incorrectly categorized positive samples, TN is the number of correctly categorized negative samples, and c is the number of categories.

TABLE II
QUANTITATIVE COMPARISON OF DIFFERENT MODELS ON THE OOWHU_CD DATASET

Methods	P(%)	R(%)	mAP_{50}(%)	$mAP_{50\sim95}$(%)
Faster RCNN	46.61	68.26	63.98	32.31
SSD	86.90	72.77	81.15	54.81
YOLOv3	80.01	73.32	77.49	54.58
M2Det	**97.62**	5.29	46.40	28.03
EfficientDet	92.52	35.10	58.21	27.03
DETR	46.90	72.26	67.98	43.90
YOLOv5	91.20	71.61	80.57	59.93
YOLOX	88.81	48.13	66.98	42.20
YOLOv7	86.82	69.81	79.43	58.58
YOLOv8	87.42	75.11	81.10	60.33
YOLOv9	86.22	**75.84**	82.05	61.12
YOLOv10	85.11	72.50	80.24	61.31
R3Det	/	/	63.32	/
S2ANet	/	/	61.88	/
ORCNN	/	/	60.44	/
SASM	/	/	70.81	/
ORepPoints	/	/	71.54	/
OoCDNet (Ours)	89.99	74.97	**82.14**	**62.99**

V. EXPERIMENTAL RESULTS AND DISCUSSION

A. Benchmark Experiments and Performance Evaluation

1) Experimental Settings: The five datasets were divided into training and validation sets at a ratio of 8:2, respectively, and the accuracy of the validation sets was used as the evaluation criterion. In the OoxBD dataset, only samples with major and destroyed levels of damage were selected. The bitemporal remote sensing images were subjected to the same "Mosaic" data enhancement operations, which involved random transformations, and the random state was frozen using the "set_random_state" method to ensure that the same random operations were performed on both images. We built OoCDNet using the Pytorch framework, and all other comparison methods are Pytorch versions except M2Det, which is a TensorFlow version. The environment for model training is based on GPU acceleration, and we used an NVIDIA GeForce RTX3090 GPU with 24 GB of memory. Adam was used as the optimizer with an initial learning rate of 0.001. The number of epochs for all experiments was 150.

2) Benchmark Comparison and Analysis: Tables II–VI show the statistics of the quantitative evaluation results of all models on the validation set of the five datasets, respectively. Figs. 8 and 9 show an example of visualization of the detection results of all models, where (#1) shows mainly large and medium objects and (#2) shows small, dense, and discretely distributed objects. From a general appreciation of the metrics, it can be observed that OoCDNet has the best overall performance because it achieved the highest or higher accuracy on all data sets. For example, for the most important metrics mAP_{50} and mAP_{50-95} of OD methods, OoCDNet's accuracy is ahead of other methods. This shows that OoCDNet's performance

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TABLE III
QUANTITATIVE COMPARISON OF DIFFERENT MODELS ON THE OOLEVIR-CD+ DATASET

Methods	P(%)	R(%)	mAP_{50}(%)	$mAP_{50\sim95}$(%)
Faster RCNN	56.58	77.40	64.73	23.22
SSD	88.10	87.29	90.41	60.24
YOLOv3	93.20	87.11	91.64	65.09
M2Det	92.44	87.79	91.81	62.02
EfficientDet	76.83	57.11	67.11	31.61
DETR	43.55	77.81	66.01	28.50
YOLOv5	93.27	87.37	92.03	66.53
YOLOX	92.77	**89.25**	92.82	64.42
YOLOv7	92.21	87.43	91.74	65.34
YOLOv8	92.31	87.00	91.82	67.02
YOLOv9	92.38	88.03	92.17	67.05
YOLOv10	90.80	88.32	92.00	**67.77**
R3Det	/	/	48.92	/
S2ANet	/	/	49.47	/
ORCNN	/	/	58.65	/
SASM	/	/	76.00	/
ORepPoints	/	/	76.34	/
OoCDNet (Ours)	**94.15**	88.29	**92.83**	67.20

TABLE IV
QUANTITATIVE COMPARISON OF DIFFERENT MODELS ON THE OOS2LOOKING DATASET

Methods	P(%)	R(%)	mAP_{50}(%)	$mAP_{50\sim95}$(%)
Faster RCNN	17.85	38.05	19.07	6.54
SSD	70.64	31.33	42.63	23.71
YOLOv3	57.24	44.81	45.48	26.93
M2Det	/	/	/	/
EfficientDet	**80.64**	6.29	29.20	13.22
DETR	18.67	50.40	26.45	12.40
YOLOv5	56.20	46.16	47.78	28.13
YOLOX	77.69	29.01	50.02	28.72
YOLOv7	58.53	46.94	51.16	31.41
YOLOv8	55.50	45.34	47.62	29.84
YOLOv9	57.78	48.19	51.58	32.19
YOLOv10	56.53	47.91	50.20	31.60
R3Det	/	/	25.66	/
S2ANet	/	/	31.30	/
ORCNN	/	/	27.27	/
SASM	/	/	28.00	/
ORepPoints	/	/	33.41	/
OoCDNet (Ours)	56.95	**52.01**	**53.08**	**32.45**

TABLE V
QUANTITATIVE COMPARISON OF DIFFERENT MODELS ON THE OOEWBD DATASET

Methods	P(%)	R(%)	mAP_{50}(%)	$mAP_{50\sim95}$(%)
Faster RCNN	24.22	32.13	18.24	5.83
SSD	75.26	22.05	42.57	18.91
YOLOv3	48.47	45.59	39.05	16.52
M2Det	**85.29**	0.26	15.77	5.91
EfficientDet	77.40	3.13	18.57	6.60
DETR	14.45	**49.79**	16.12	5.00
YOLOv5	50.98	45.60	42.48	18.75
YOLOX	71.70	25.75	43.70	18.57
YOLOv7	49.72	48.60	44.50	19.50
YOLOv8	50.10	46.33	43.91	20.10
YOLOv9	50.60	48.14	44.90	20.12
YOLOv10	48.11	45.23	42.38	19.52
R3Det	/	/	45.35	/
S2ANet	/	/	45.18	/
ORCNN	/	/	43.05	/
SASM	/	/	45.19	/
ORepPoints	/	/	45.28	/
OoCDNet (Ours)	50.03	47.74	**46.43**	**20.40**

TABLE VI
QUANTITATIVE COMPARISON OF DIFFERENT MODELS ON THE OOXBD DATASET

Methods	P(%)	R(%)	mAP_{50}(%)	$mAP_{50\sim95}$(%)
Faster RCNN	33.76	**67.77**	49.51	20.74
SSD	76.71	52.78	63.58	34.02
YOLOv3	79.02	42.00	57.11	29.04
M2Det	**88.44**	13.25	45.21	20.09
EfficientDet	27.41	65.32	47.59	19.92
DETR	69.99	62.22	65.90	37.20
YOLOv5	72.28	62.53	67.42	38.56
YOLOX	82.13	54.77	68.72	38.07
YOLOv7	71.25	62.59	67.15	36.34
YOLOv8	71.43	61.19	67.92	40.33
YOLOv9	68.44	63.91	68.70	41.32
YOLOv10	71.73	61.20	68.02	38.07
R3Det	/	/	44.43	/
S2ANet	/	/	46.72	/
ORCNN	/	/	46.33	/
SASM	/	/	47.93	/
ORepPoints	/	/	52.83	/
OoCDNet (Ours)	74.84	65.54	**71.70**	**42.82**

in OD is extraordinary. In other words, OoCDNet's ability to detect changing or damaged building objects is outstanding, and it has significant superiority over other methods. In terms of P and R, OoCDNet does not take the complete lead, though. However, OoCDNet strikes a balance between accuracy and reliability. This is because, although the P-value of some methods is high, the R-value is low, which indicates that the misdetection of these models is high. It is also easy to see from Figs. 8 and 9 that OoCDNet has the most accurate detection of objects on all datasets, and it has low false and missing detections. In contrast, the other models have high false detections and missed detections, and some models did not even detect the changed buildings. At the same time, OoCDNet has good adaptability to large, medium, and small objects, and the detection boxes always fit the buildings nearly perfectly. Although models such as ORepPoints attempt to make the bounding boxes closer to the target through rotation, they do not perform well for this CD task based on bitemporal imagery. A significant amount of missed detections and false detections affect their accuracy. For example, the rotated bounding boxes generated by S2ANet have noticeably distinct angles, even generating rotated boxes

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Fig. 8. Visual comparison of the results of the basic experiments.

for small buildings. While these rotated boxes may sometimes fit the target more closely, they do not necessarily contribute to more accurately locating the target. For small buildings, the benefit of using rotated boxes is limited, especially compared

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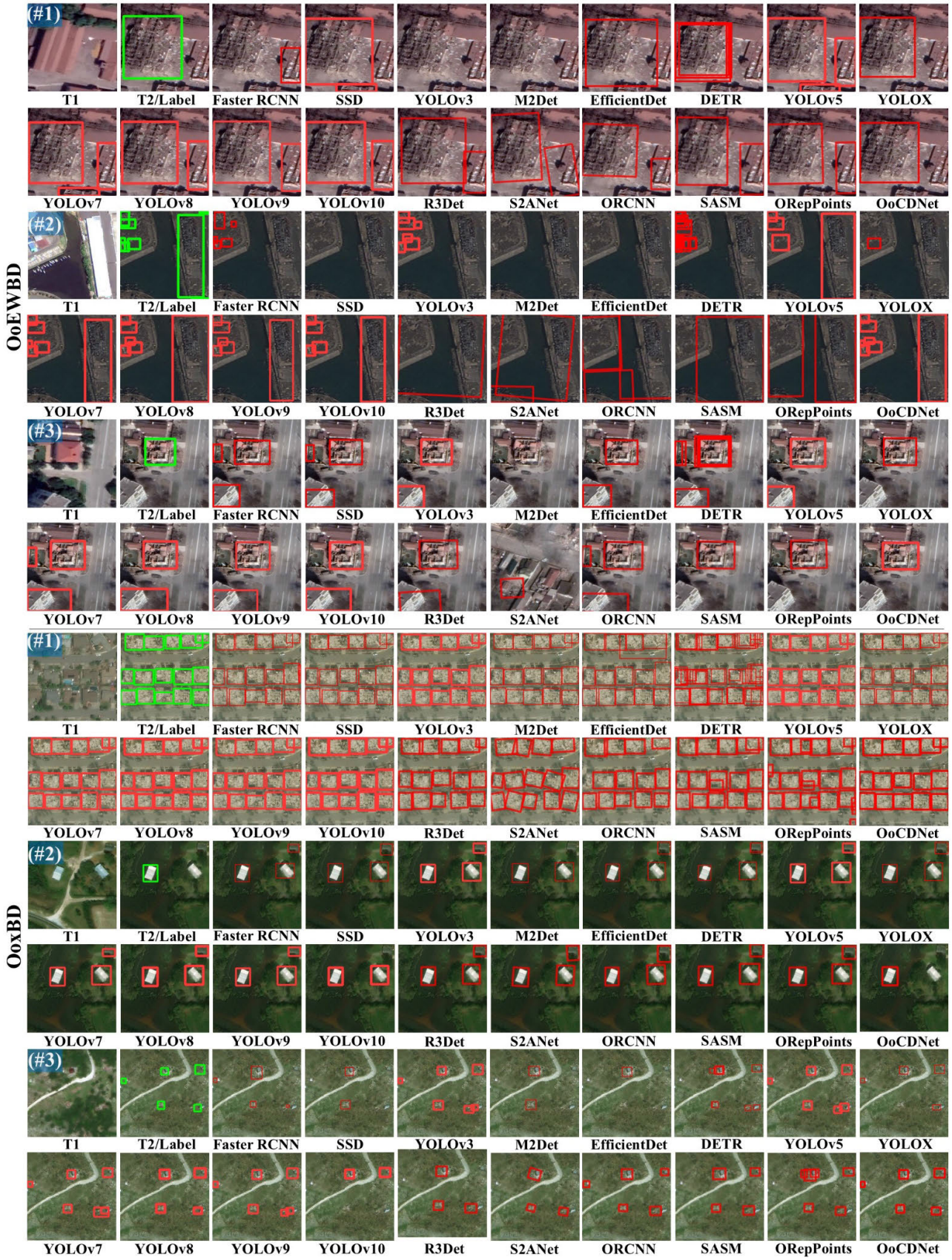


Fig. 9. Visual comparison of the results of the basic experiments (OoEWBD and OoxBD datasets).

to the significant number of missed detections. In terms of the models themselves, SSD, YOLOv3, YOLOv5, YOLOX, and YOLOv7 perform a little better than the other methods, which we believe may be related to the structure of the models.

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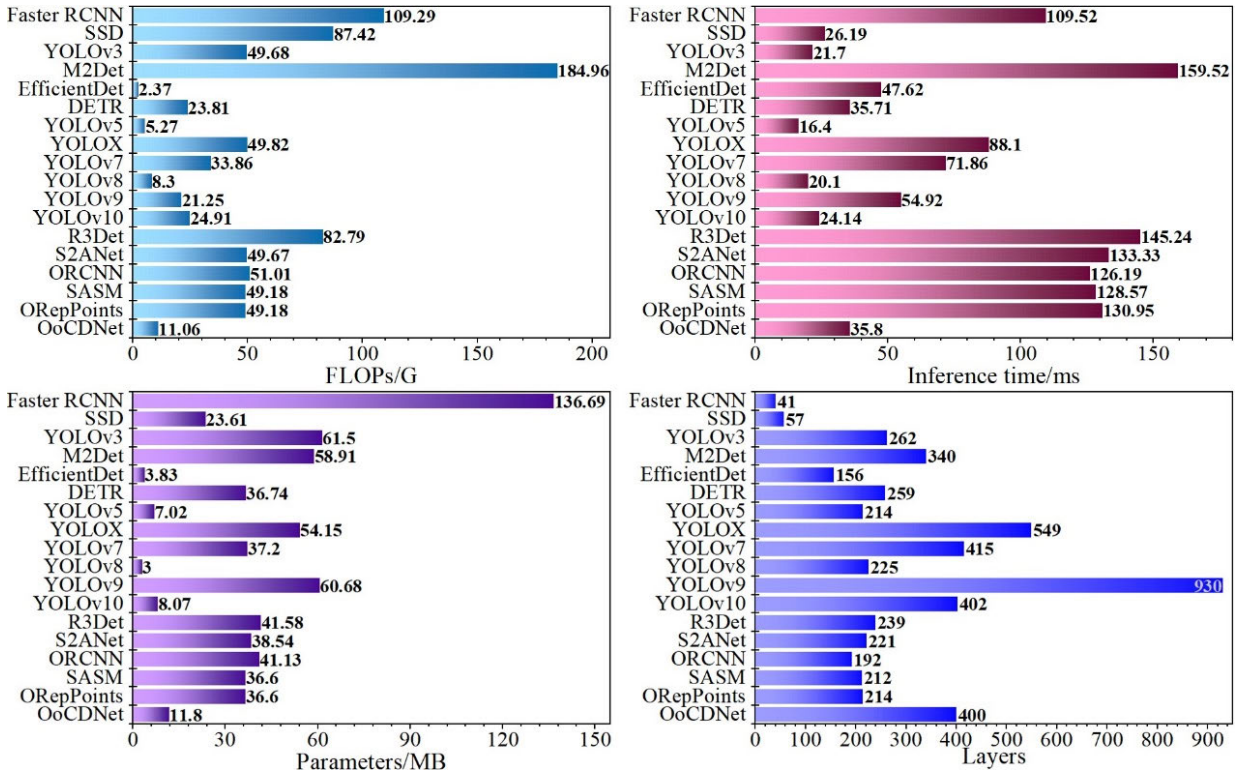


Fig. 10. Comparison of OD-based models.

For example, although DETR utilizes the transformer encoding and decoding structure, the model does not accomplish good convergence in 150 epochs. The transformer mechanism, which requires a large number of training samples and training epochs, may not be suitable for OD-based BCD tasks.

M2Det even shows unfit on some datasets; it cannot be denied that it performs well for specific OD tasks on normal images, but for objects, such as changed buildings, M2Det may not be suitable. EfficientDet has a good feature fusion strategy, but OD-based BCD tasks need to capture the changing features efficiently. It may be difficult to directly transfer it to bitemporal imagery. It may be difficult to directly transfer a strategy that is effective for normal images to bitemporal imagery. The structure of SSD is not complicated, without too many tricks and strategies, perhaps because this direct extraction of features makes it obtain higher accuracy. In addition, the YOLO series of networks all achieved higher accuracy, which is related to the overall idea of this series of networks. Especially the recently released models YOLOv8, YOLOv9, and YOLOv10, all of which have high accuracy and can even match the performance of OoCDNet. However, their performance on the OoS2Looking and OoEWBD datasets is average, indicating that it is difficult to capture changes or damage characteristics between bitemporal images using only a single encoding path. It is worth mentioning that Faster RCNN showed moderate and stable performance on all datasets. Its mode of operation may be the key to endowing it with this capability. The OoCDNet dual-path feature extraction structure, as well as the self-attentive information enhancement structure, which is designed to target OD-based BCD tasks, is the basis for its leading performance.

All models perform better on OoWHU_CD and OoLEVIR-CD+ than on the other two datasets. This can also be visualized in Figs. 8 and 9, where, for the OoLEVIR-CD+ dataset, where the change features are obvious, the scene is not too complex, and the amount of data is more than sufficient, all the models achieve convergence, with near-perfect detection results obtained for OoCDNet, YOLOv10, YOLOv9, YOLOv8, YOLOv7, YOLOX, and YOLOv5. This proves that the change of buildings from nothing or from something to nothing is regular, which is easy for the model to learn. The regular changes are reflected in the data level as more easily fit distribution states.

On the contrary, the images in the OoS2Looking dataset were taken from multiple angles, and there are more complex scene variations between the bitemporal images. In the OoxBD dataset, although many buildings are defined as major damaged or destroyed, some of them are only affected by flood imagery, with no actual changes to the buildings themselves. The damaged buildings in the OoEWBD dataset, on the other hand, are more diverse in terms of spectra, textures, and so on. It is difficult for the models to feature extract and learn the OD-based change relationship. mAP_{50} of many models did not exceed 30%, and some models even failed. Although methods such as R3Det, S2ANet, ORCNN, SASM, and ORepPoints are designed for remote sensing images, their performance is often unsatisfactory. They perform even worse than YOLOv3 on the OoWHU_CD and OoLEVIR-CD+ datasets. This may also indicate that models based on oriented bounding boxes

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TABLE VII
ACCURACY COMPARISON OF PATH ABLATION EXPERIMENTS

Methods	OoWHU_CD		OoLEVIR-CD+		OoS2Looking		OoEWBD		OoxBD	
	single	dual	single	dual	single	dual	single	dual	single	dual
YOLOv3	77.49	80.97↑	91.64	91.72↑	45.48	45.60↑	39.05	42.19↑	65.90	67.79↑
YOLOv5	80.57	80.22↓	92.03	92.45↑	47.78	48.81↑	42.48	45.88↑	67.42	68.16↑
YOLOv7	79.43	80.47↑	91.74	91.77↑	51.16	51.84↑	44.50	44.30↓	67.15	68.46↑
OoCDNet	81.47↓	82.14	92.73↓	92.83	50.11↓	53.08	46.00↓	46.43	70.150↓	71.70

Note: The values of the statistics in the table are mAP_{50} (%), the arrows indicate the increase or decrease, single indicates a single-path structure, and dual indicates a two-path structure.

are difficult to adapt to tasks involving horizontal bounding boxes. However, OoCDNet maintains the highest mAP_{50} value and is significantly ahead of most of the methods. On the OoS2Looking dataset, OoCDNet even outperforms the second place by nearly 3%. Furthermore, discretely distributed objects and objects in complex scenes are difficult to detect [e.g., Fig. 8, the first image (#2), and Fig. 9 (#1)]. It is clear from Figs. 8 and 9 that OoCDNet accurately discovers such objects in all cases, and the detection boxes are of moderate size, just enclosing the areas of the buildings that have been changed or damaged. This may prove that OoCDNet's CSCEBlock effectively captures key change information at the semantic level and improves the accuracy of the detection boxes with low-dimensional localization information, which makes OoCDNet excel in both the accuracy of the detection boxes, as well as the change feature extraction in complex scenes.

To further evaluate the overall complexity, model volume, and efficiency of OoCDNet, all models were statistically analyzed in terms of FLOPs, number of parameters, number of network layers, and average inference time. Among them, the average inference time is calculated based on the OoWHU_CD validation set. Fig. 10 shows the statistical results. From the figure, it can be observed that the FLOPs' value of OoCDNet is less than 14 out of 17 comparison methods, which indicates that its computational effort is small and the model can be trained in a shorter time. Moreover, OoCDNet's inference is also faster, taking less time than most comparison models. This is crucial for quickly completing detection and outputting results in emergency situations, making it well-suited to support urgent disaster relief efforts. Moreover, the inference speed of the five OD methods based on remote sensing images, such as R3Det, is very slow, which greatly limits their real-time application. Compared to YOLOv7, YOLOv8, YOLOv9, and YOLOv10, which performs better in terms of accuracy, OoCDNet achieves a tradeoff between both accuracy and model volume. It is worth mentioning that although OoCDNet has more layers, its parameter count is very small and not in the same volume range as most models. This is friendly to the onboard deployment of intelligent remote sensing satellites, which have limited storage space after all. Taken together, OoCDNet achieves a balance in terms of model accuracy, efficiency, and volume.

B. Discussion

1) Ablation Study:

a) Dual-path structural ablation experiment: We believe that the OD-based BCD method can achieve the compatibility of both CD and OD methods, which is first related to the structure of the network, i.e., the images are fed into the network from two branches separately, which will extract more effective object change features. The common OD task focuses on extracting object features from RGB images, whereas the BCD task focuses on the changing features of the object. Inputting bitemporal images through one path is difficult for the model to understand and model. To analyze the impact of the dual-path structure on the OD-based BCD method, we have constructed the single- and dual-path structures of YOLOv3, YOLOv5, YOLOv7, and OoCDNet based on their four networks. Among them, the single-path results for YOLOv3, YOLOv5, and YOLOv7 and the dual-path results for OoCDNet have been obtained in Section V-A2. We retrained them on the four datasets using the newly constructed YOLOv3, YOLOv5, and YOLOv7 dual-path networks, as well as the OoCDNet single-path network. The mAP_{50} values shown in the Table VII certainly confirm our suspicions. When the model was expanded from a single path to a dual path, the accuracy was improved. In contrast, when OoCDNet is compressed from a dual path to a single path, the accuracy decreases. This suggests that the model structure of the dual path is crucial for the OD-based BCD method.

b) CSCEBlock ablation experiment: The CSCEBlock is a key component in OoCDNet for capturing critical change information and performing information enhancement, and we designed ablation experiments to analyze its value and usefulness. First, we removed CSCEBlock from the model, replaced it with ordinary skip connections, and reconstructed the model, notated as OoCDNet-C. Following the same experimental conditions, we trained OoCDNet-C on four datasets and counted its metrics. The results are shown in Fig. 11.

It is easy to find that CSCEBlock can help the model focus the attention on the changed buildings. In contrast, after removing CSCEBlock, the attention is disoriented, the model finds it difficult to focus on the key areas in the image, and there are positional deviations and missing phenomena occurring. OoCDNe also achieves attentional focus on small objects that are not easy to spot, which may be related to the strong localization information that CSCEBlock gives to the model. It can also be noticed from Fig. 11 that OoCDNe can focus its attention on the center of the target, which indicates that CSCEBlock is adapted for the OD-based BCD task. Because the OD-based BCD task is oriented to discovering the

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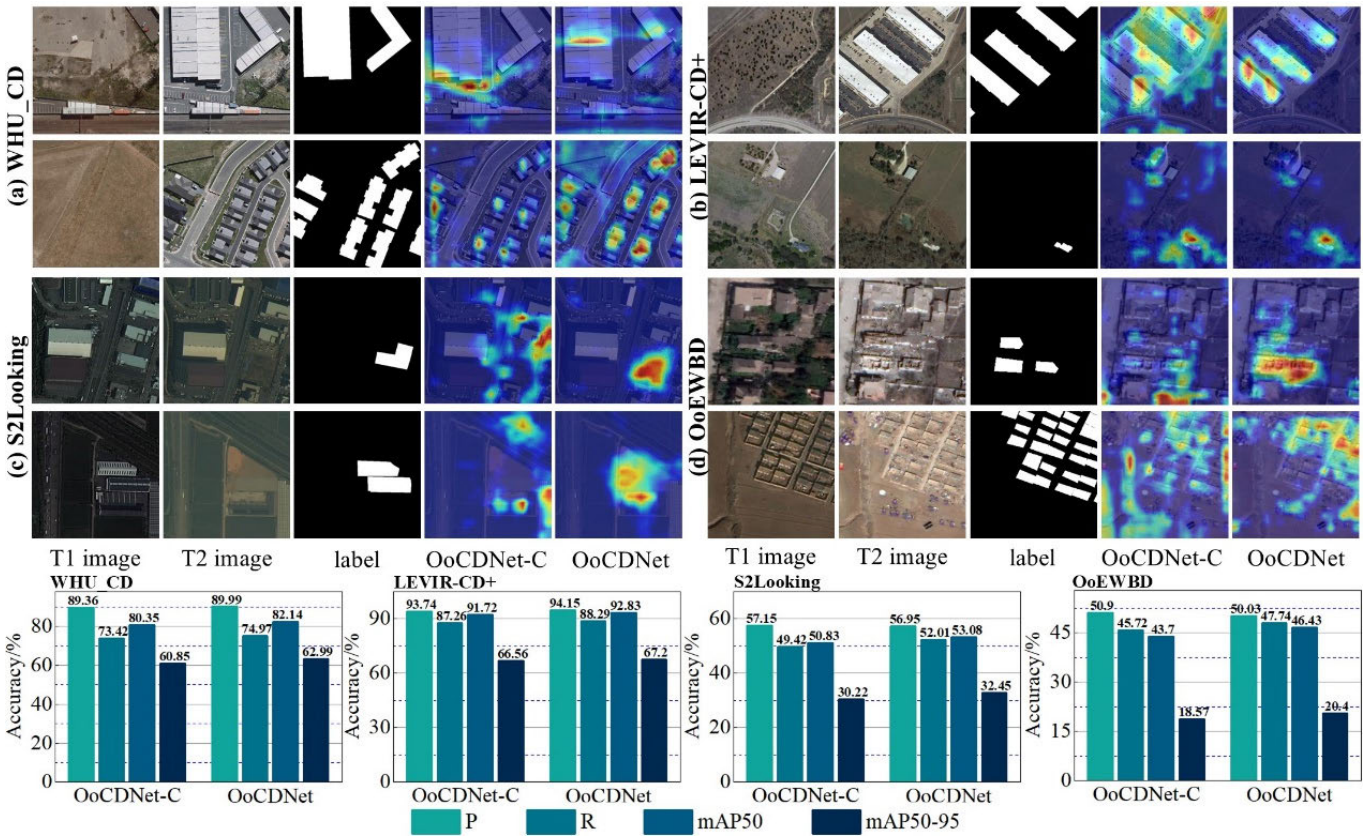


Fig. 11. CSCEBlock ablation experiment accuracy statistics and attention visualization.

position of a changing object, this precise focusing of attention is valuable for detecting the position of the object. The bottom row in Fig. 11 shows the change in accuracy before and after the removal of CSCEBlock. The accuracy of the model is significantly reduced after CSCEBlock is removed, with nearly 2% attenuation in almost all metrics, and the magnitude of this attenuation is larger than the performance of the model when it is compressed into a single path.

2) Comparison With CD-Based Methods:

a) Accuracy comparison: To further evaluate the advantages of OD-based BCD methods, we conducted generalized BCD experiments based on three publicly available datasets, namely, WHU_CD, LEVIR-CD+, and S2Looking. We selected the commonly used segmentation models, UNet, FC_EF_Res, DeepLabv3+, and BIT, as well as two object-based methods to compare with OoCDNet. It should be noted that the aforementioned four segmentation models are also used as comparison methods in many DP-based BCD studies. The comparison with object-based methods is intended to evaluate the performance of OoCDNet in object-level CD. We trained the model for 150 epochs based on the same three datasets mentioned above. The data used for training and validation are the same as those used for conducting OD-based experiments. Fig. 12 shows the accuracy of the DP-based BCD model on the validation set. It can be found that the OoCDNet object-level P, R, and mAP_{50} values even exceed the DP-based BCD model pixel-level metrics on the three datasets. It is worth noting that the OD-based metrics are computed in a way related to the confidence threshold and the

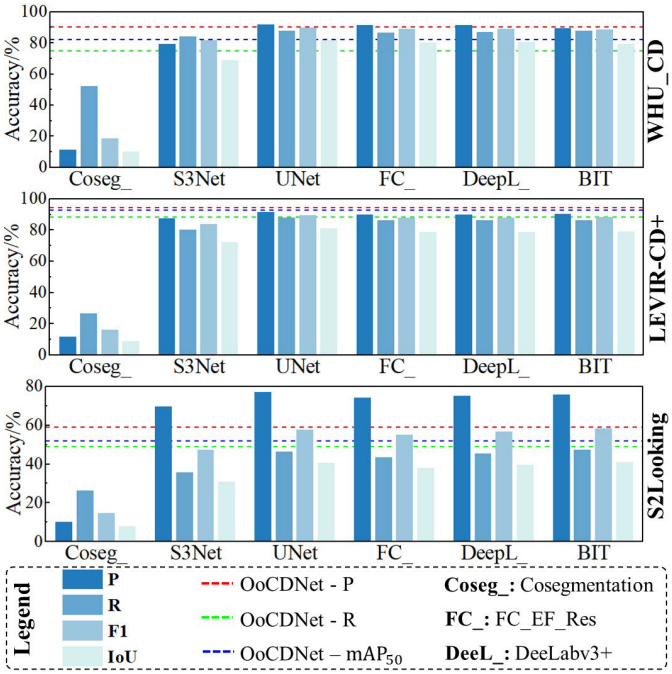


Fig. 12. Comparative analysis of the accuracy with the DP-based BCD method.

IoU threshold. When we calculated the metrics for OoCDNet, the two metrics were chosen to be 0.001 and 0.6.

The precision values obtained in this case do not fully reflect the detection results of the model, in contrast to the

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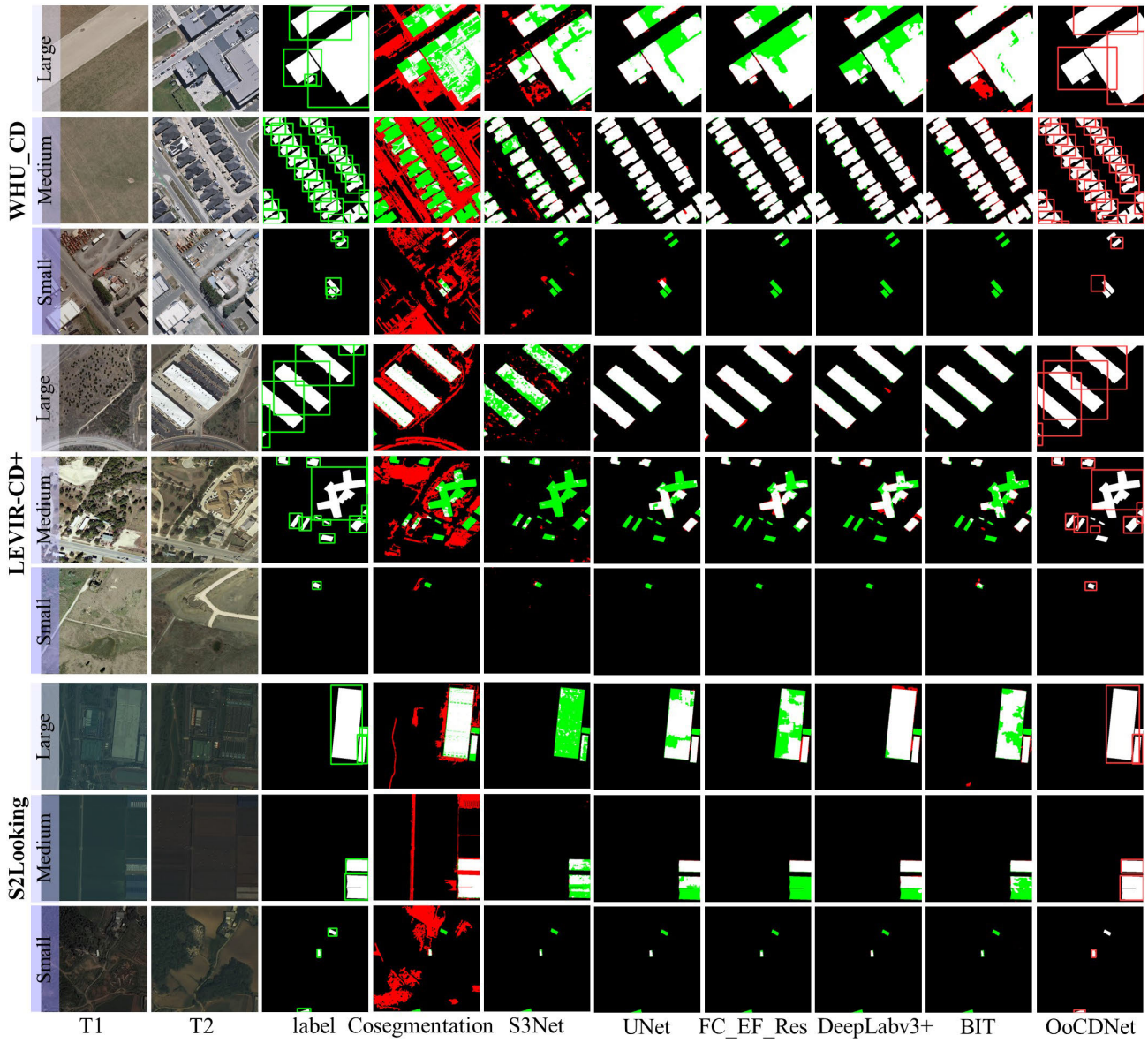


Fig. 13. Visual comparison of detection results between OD- and DP-based BCD methods. The rendered colors represent TPs (white), FPs (red), FNs (green), and TNs (black).

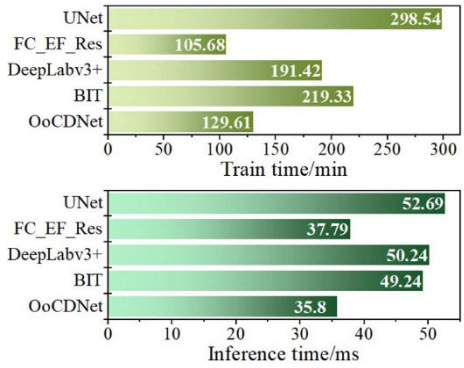


Fig. 14. Comparison of efficiency between OD- and DP-based BCD methods.

DP-based CD method, which fully counts all pixels. Therefore, it is reasonable to believe that OoCDNet surpasses the general BCD model in terms of accuracy. It can also be shown that our proposed new OD-based BCD model can perform object-level BCD tasks.

Fig. 13 shows a visual comparison between the DP-based BCD method and OoCDNet, and we have chosen three types of objects, large, medium, and small, for detailed analysis. It can be seen that OoCDNet can accurately detect the various types of objects that have changed and have fewer misses and false detections. In contrast, the DP-based BCD method is more effective in detecting large- and medium-sized targets, but the phenomenon of missed detection occurs commonly. Especially for small objects, the DP-based BCD method is characterized by serious missed detections. The performance of the two object-based methods is poor, with a significant amount of missed detections and false detections. This suggests that OoCDNet, in addition to its general BCD application value, is highly robust to small changes that are not easily detected.

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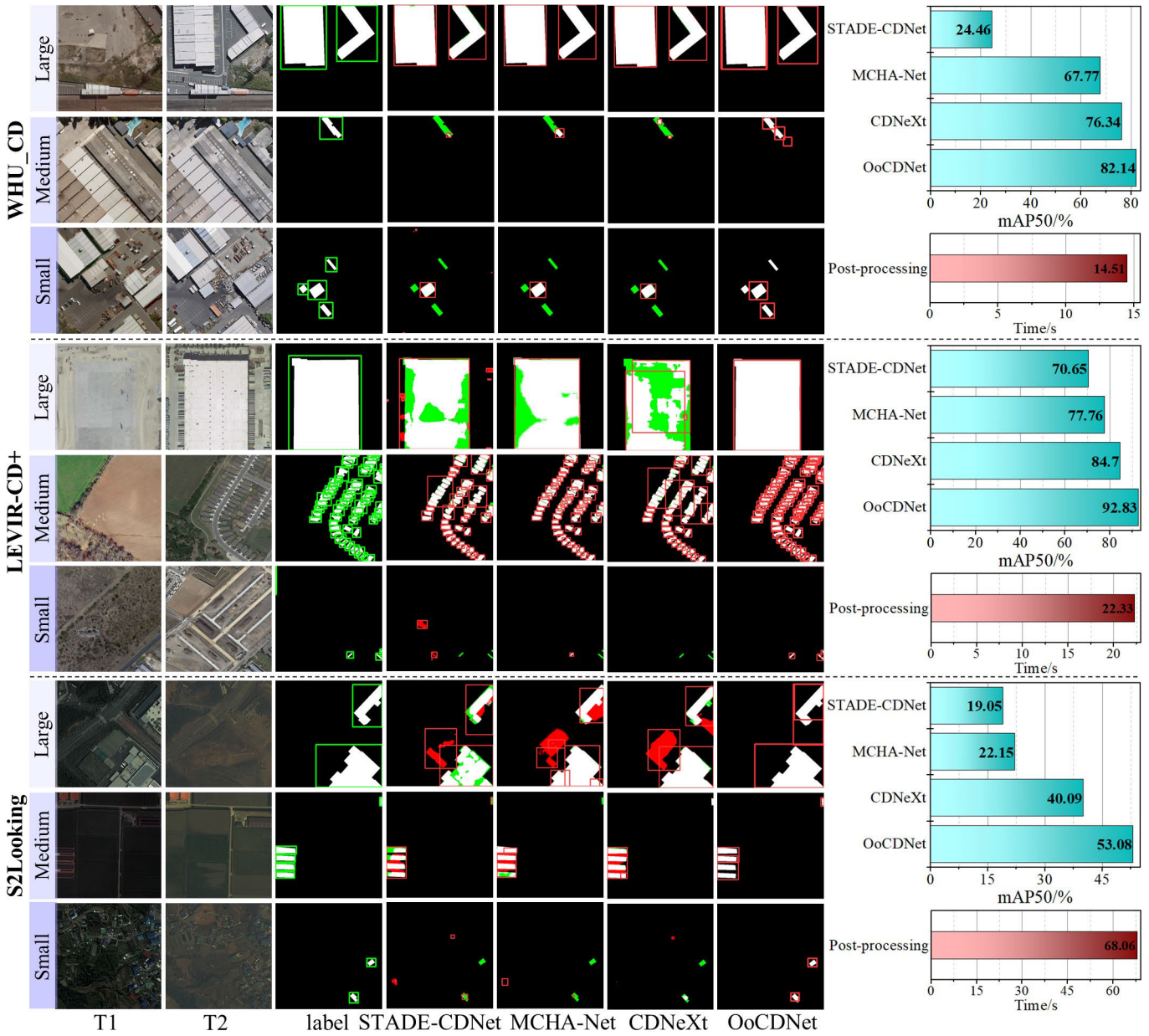


Fig. 15. Comparison with SOTA CD networks based on postprocessing techniques. The rendered colors represent TPs (white), FPs (red), FNs (green), and TNs (black).

b) Efficiency comparison: To demonstrate the efficiency of our proposed OD-based BCD working model, we counted the time consuming of the four DP-based BCD methods and OoCDNet to complete the training of 150 epochs on the WHU_CD training set and to complete the inference on each patch on the validation set.

As shown in Fig. 14, it can be found that the training time of OoCDNet is significantly smaller than that of the DP-based BCD method, except that it takes slightly longer than FC_EF_Res. Moreover, the time consumption of OoCDNet is almost times smaller than the DP-based BCD method. This indicates that the practical application value of OoCDNet is significant because the training time consumption of deep learning models has always been an important factor restricting its application. In terms of inference time, OoCDNet gets the best performance. It takes almost 30 ms less time on average than UNet, which is known to be almost the most basic segmentation model, and most DP-based BCD models are almost more complex than it. To be sure, OoCDNet is valuable to be applied in some emergencies. For example, when an earthquake or an explosion occurs in a certain place and it is difficult for rescuers to enter, the use of OoCDNet can provide a fast census of damaged houses based on remote sensing images for emergency rescue. Furthermore, this fast inference, as well as the lightweight of the model itself, is friendly for on-planet deployment. It is highly realistic to realize the on-planet deployment of OoCDNet with real-time detection of changed/damaged houses in the mission area.

3) Comparison With State-of-the-Art CD Networks Based on Postprocessing Techniques: Pixel-level state-of-the-art (SOTA) CD networks can achieve high-precision detection

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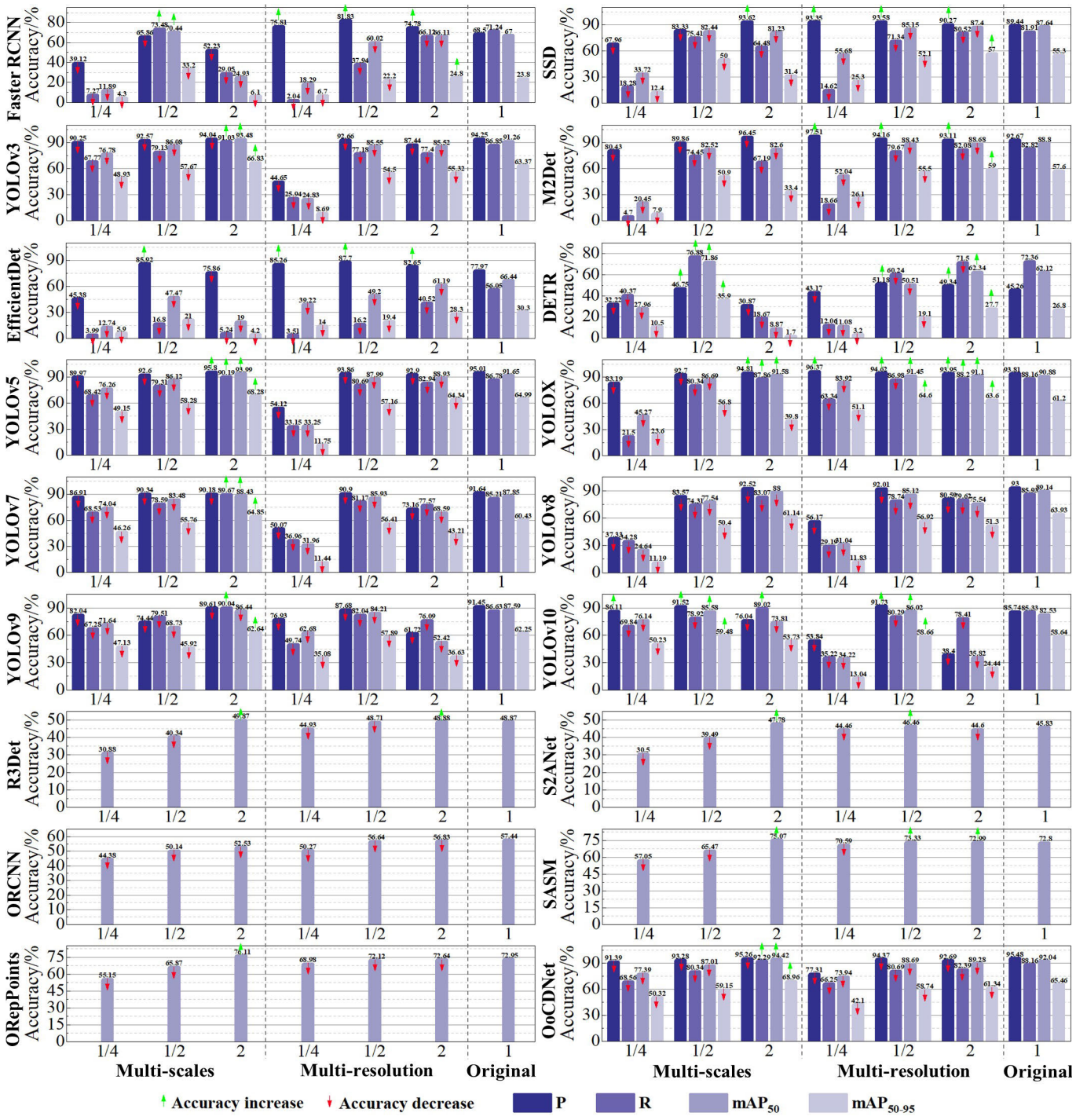


Fig. 16. Model accuracy at multiple scales and resolutions.

results. By applying postprocessing techniques to these detection results, similar outputs to OoCDNet can be obtained. To analyze the differences between OoCDNet and SOTA CD networks, as well as to demonstrate the superiority of OoCDNet in handling OD-based CD tasks, we designed a comparative experiment between OoCDNet and SOTA CD networks. We selected three latest CD networks, CDNeXt [94], MCHA-Net [27], and STADE-CDNet [95], whose authors have sufficiently demonstrated their performance in DP-based CD through extensive experiments.

We retrained these three models on three different CD datasets and conducted inference on the test set data under the same experimental conditions. We utilized the "findContours" and "boundingRect" functions from OpenCV to design a postprocessing program that can obtain the corner coordinates of each building instance. Based on this postprocessing program, we mapped the OD results to the inferred binary images. Finally, we visualized the mapping results onto the binary images and calculated the mAP_{50} value based on the labels. Fig. 15 provides a detailed statistical summary of

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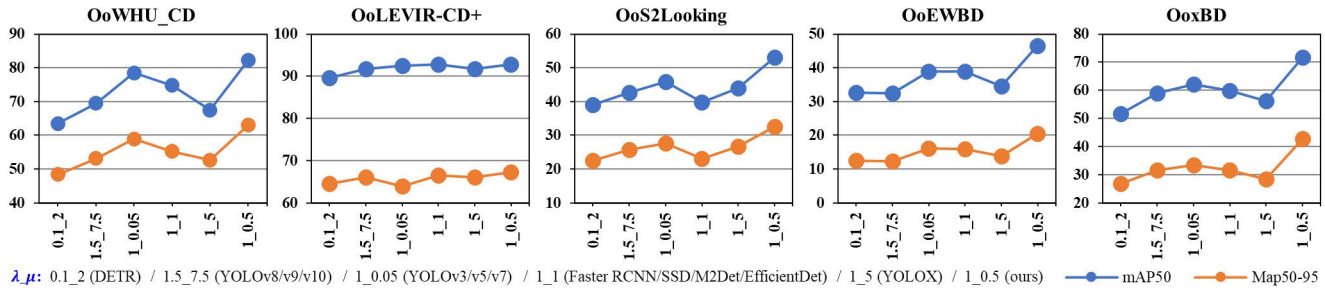


Fig. 17. Discussion on the configuration schemes for λ and μ.

each network's prediction results for each dataset, along with quantitative evaluations.

It can be observed that the mAP_{50} values of the three SOTA CD networks differ significantly from those of OoCDNet, with differences exceeding 10% on the LEVIR-CD+ and S2Looking datasets. While their pixel-level F1 scores might exceed 90%, they do not achieve satisfactory mAP_{50} values in OD tasks. This is related to the false positives and missed detections of SOTA CD networks. Discrete speckle noise significantly increases the false detection rate, and the incompleteness of large target detection results also raises the number of false detection objects. Furthermore, densely connected targets with unclear boundaries lead to a significant increase in missed detections because multiple objects are incorrectly detected as one. Although the inference results of SOTA CD networks can achieve outputs similar to OoCDNet through postprocessing techniques, the time consumption of the postprocessing process cannot be ignored. As shown in the statistical results on the right-hand side of Fig. 15, the postprocessing process can take more than a minute, which is disadvantageous for real-time applications in some emergency fields. In addition, considering the high training time costs of these SOTA CD networks, their overall time cost becomes even higher.

4) Performance Evaluation at Multiple Scales and Resolutions: The remote sensing imagery conditions involved in conducting urban development assessments or emergency analysis work are often diverse and complex. For example, the assessment of changes in urban sequences may require the use of images of different resolutions. Rapid mapping after disasters such as earthquakes and explosions may have to be performed based on imagery of very large or very small areas. In this case, the stability of the model's performance in multiscale and multiresolution situations is extremely important for the work to be carried out.

To verify the performance of OoCDNet under multiscale and multiresolution conditions, we constructed a validation dataset with resolutions and scales of 1/4, 1/2, and 2 times the original image based on part of the OoLEVIR-CD+ dataset. The accuracy is evaluated on the validation dataset using the model in Section V-A2. As shown in Fig. 16, the accuracy of all models on the validation dataset fluctuates. However, OoCDNet has the best stability. Its accuracy under other conditions fluctuated the least relative to the original accuracy, and it achieved the maintenance of higher accuracy except for a more subtle decrease in accuracy on the 1/4 resolution dataset. In contrast, YOLOX and the five rotated OD methods exhibit better stability. All other models show different degrees of performance degradation. We believe that this is related to the overall structure of OoCDNet and the attention mechanism. It can also be demonstrated that OoCDNet can be applied under complex imaging conditions and can be used as a technical tool for urban development assessment and emergency response.

5) Analysis of the Values of λ and μ: To further discuss the impact of the values of λ and μ on the model, we set up five configurations of λ and μ based on the configuration parameters of $Loss_{obj}$, $Loss_{box}$, and $Loss_{cla}$ (classification loss) in 12 OD networks. We then retrained OoCDNet on five datasets using these five configurations. After training, we plotted Fig. 17 based on their quantitative performance on the test sets. It can be seen that when λ and μ are set to 1 and 0.5, respectively, OoCDNet achieves the highest accuracy. In other cases, the model's accuracy shows varying degrees of degradation. Notably, when the value of μ exceeds that of λ, the accuracy drops significantly. However, when λ and μ are set to 1 and 0.05 or 1 and 1, the model achieves relatively high accuracy. This indicates that in the OD task of damaged/changed detection in buildings based on bitemporal remote sensing images, the proportion of bounding box loss in the overall loss should be increased, but the dominance of confidence loss must be maintained. For OD tasks that do not require classification, it might be beneficial to slightly increase the bounding box loss while emphasizing confidence loss to achieve a balanced tradeoff between precision and recall.

6) Large-Scale Generalizability Tests: Furthermore, to analyze the generalization ability of OoCDNet on large-scale scenarios, we evaluate the model's capability for census and mapping of changing buildings, as well as its potential for application in urban development assessment. We tested OoCDNet on the whole Mts-WH dataset [42] and the WHU_CD dataset. In particular, the area covered by the images in the Mts-WH dataset is the Hanyang district of Wuhan, China, and the images were acquired in February 2002 and June 2009, respectively. During this period, the city was under rapid construction, with a large number of residential, commercial, or industrial buildings being built. The dataset contains two remote sensing images of 7200 × 6000 pixels with 1-m resolution and corresponding DP-based labels. The image coverage in the WHU_CD dataset is of

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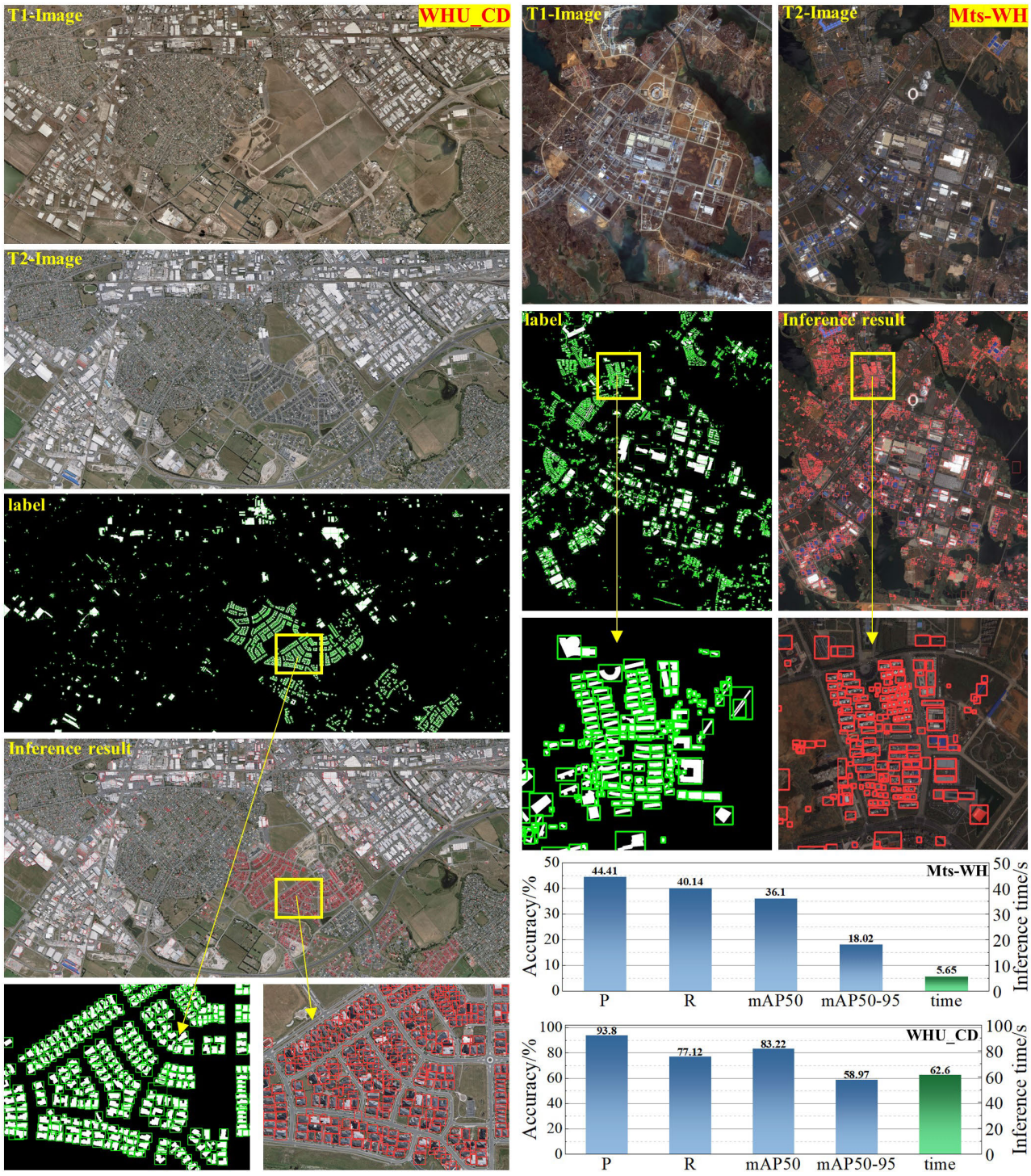


Fig. 18. OoCDNet generalization ability test under large-scale images.

the city of Christchurch, New Zealand, where a magnitude 6.3 earthquake occurred in February 2011, destroying a large number of buildings. A large number of buildings were rebuilt in the following years. Two images of size 15 354 × 32 507 pixels with a resolution of 0.075 m were acquired for the periods 2012 and 2016, respectively. We labeled OD-based labels based on DP-based labels. It should be noted that inference on Mts-WH is performed using the model obtained by training on the S2Looking dataset. In addition, we also counted the inference time consumption and evaluated the accuracy of the inference results. As shown in Fig. 18, without model fine-tuning, OoCNDet can obtain more than 35% of mAP_{50} values on the Mts-WH dataset and takes less than 6 s to complete the generation of inference results for large-size images. OoCNDet even achieves more than 83% mAP_{50} value on the WHU_CD dataset. It can also be seen from the visualization results that OoCDNet achieves the detection and localization of changing building objects in a large area.

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Fig. 19. OoCDNet model generalization ability test results.

Comprehensively, it shows that OoCDNet can satisfy the census of changing buildings in large-scale scenarios and can support the urban development assessment work. It can also prove that this new OD-based BCD model that we designed has strong practical value.

In addition, the models trained on the four datasets, excluding OoS2Looking, achieve mAP_{50} values ranging from 20% to 35% on the Mts-WH dataset. This indicates that factors such as the complexity and learning difficulty of images in the training datasets directly affect model performance. Images in OoS2Looking have more challenging change mapping relationships. When models reach a fitting state on such datasets, they tend to perform better on more easily distinguishable image changes and generally exhibit stronger

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TABLE VIII
COMPREHENSIVE PERFORMANCE ANALYSIS OF MODELS

Methods	FLOPs(G)	Train time (hour)	Inference time(s)	mAP_{50}(%)
Faster RCNN	109.29	14.13	18.07	32.47
SSD	87.42	7.82	4.32	34.47
YOLOv3	49.68	5.97	3.58	39.62
M2Det	184.96	10.01	26.32	32.49
EfficientDet	2.37	6.01	7.86	37.94
DETR	23.81	5.72	5.89	34.54
YOLOv5	5.27	4.82	2.71	40.35
YOLOX	49.82	4.27	14.54	39.21
YOLOv7	33.86	10.18	11.86	31.22
YOLOv8	8.30	10.43	3.32	38.55
YOLOv9	21.25	4.38	9.06	41.63
YOLOv10	24.91	5.12	3.98	42.83
R3Det	82.79	3.38	23.96	43.08
S2ANet	49.67	3.98	21.99	39.47
ORCNN	51.01	4.20	20.82	40.18
SASM	49.18	5.08	21.21	42.42
ORepPoints	49.18	6.30	21.61	45.37
OoCDNet (Ours)	11.06	5.12	5.91	49.52

detection performance. This suggests that models trained on datasets with higher image complexity often demonstrate greater robustness.

Furthermore, we used images of the Lahaina area taken by the Jilin-1 03D28 satellite on August 13, five days after the Maui, HI, fire, as well as images from Google Earth on March 24, 2023, for OD-based BCD and mapping of damaged or destroyed buildings. The model used was obtained by training on the OoEWBD dataset. The resolution of the images was uniformly resampled to 0.75 m. Fig. 19 shows the detection and mapping results obtained by the OD-based BCD method. From (a), it can be seen that the damaged houses in the Lahaina area are mostly concentrated, mainly in the north, south, and east, which is consistent with the manually annotated results. This indicates that the OoCDNet model obtained from the training based on the OoEWBD dataset is more sensitive to the building-damaged features, and the model generalization ability is stronger, which can detect damaged buildings in a wide range of unfamiliar areas.

Similarly, models trained on the four datasets, excluding OoEWBD, performed poorly on Google Earth imagery from the Lahaina region, with mAP_{50} values generally falling below 40% (compared to 49.52% for OoEWBD). First, a significant portion of the OoEWBD dataset is sourced from Google Earth, which introduces a similarity between the training data and the test data. This homogeneity enables the model to learn more similar mapping relationships. Second, since OoEWBD focuses primarily on damaged buildings, the models trained on this dataset are more sensitive to damage features. When detecting buildings with similar damage in the Lahaina region, the model's sensitivity to these features allows it to identify and accurately recognize a larger number of objects. Therefore, it is reasonable to conclude that OoEWBD is a necessary condition for the strong generalization capability of OoCDNet.

To comprehensively analyze the complexity and efficiency of the model itself, we evaluate the overall effectiveness of OoCDNet when applied to emergency services. Based on the performance of the model trained on the OoEWBD dataset and tested on Hawaiian data, we gathered statistics on the FLOPs, training time, inference time, and mAP_{50} for 18 different methods. The training time refers to the total duration for 150 epochs on the OoEWBD dataset, while the inference time is the total duration for predicting the Hawaiian images. mAP_{50} reflects the accuracy of the model on the Hawaiian images. Table VIII summarizes these four metrics for all models. It is evident that OoCDNet yields the greatest benefits, with lower model complexity and high efficiency. With an inference time of only about 5 s, it demonstrates significant potential in disaster response scenarios. The Hawaiian images are 5328 $\times$ 7574 pixels in size, and OoCDNet's ability to perform fast and accurate inference on such large-scale images indicates its important role in emergency services. Moreover, its mAP_{50} on unfamiliar images reaches 49.52%. Even if model fine-tuning is required, OoCDNet remains highly efficient, completing training on the OoEWBD dataset of approximately 10 000 training and test samples in about 5 h. In contrast, the overall performance of other models is almost always inferior to OoCDNet.

VI. CONCLUSION AND FUTURE WORK

Rapid detection and mapping of changed buildings using bitemporal HR remote sensing imagery in multiple scenarios around the world in the context of natural and man-made disasters are highly significant but challenging tasks for urban development assessment, emergency response, and humanitarian assistance. This study redefines a new model of BCD that breaks through the traditional way of thinking about CD. The technical advantages and workflow are integrated from both OD and CD perspectives. We have designed a network model, OoCDNet, capable of performing rapid detection of changed buildings. To support the development of OD-based BCD work and subsequent research, we constructed OoEWEBD, a bitemporal damage building dataset based on a global scale, different causes, different scenarios, and different data sources, as well as the change/disaster-affected building datasets, OoWHU_CD, OoLEVIR-CD+, OoS2Looking, and OoxBD. Extensive experiments confirm the feasibility of our proposed OD-based BCD working model and the superiority of OoCDNet performance. Ablation and comparison experiments demonstrate that OoCDNet is well-structured and the CSCEBlock attention mechanism is effective. Compared with ordinary CD methods, OoCDNet performs amazingly well in terms of accuracy, efficiency, and robustness. Model testing based on large-scale imagery proves that OoCDNet has strong robustness and generalization ability to support the detection of changed buildings under unfamiliar imagery conditions.

In future research, we will further optimize the workflow of OD-based BCD and work on improving the accuracy and efficiency of OoCDNet, designing a more lightweight

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version of the model and implementing OoCDNet for satellite-based deployment. For example, implementing the OoCDNet algorithm for real-time on-orbit detection on the LuoJia3-01/02 intelligent remote sensing satellites. In addition, we need to further enrich the OoEWEBD dataset to continuously improve the fundamental value of OoEWEBD in damaged building detection by adding more sensors and more remote sensing images in more scenarios.

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